



TEHNIČKO VELEUČILIŠTE
U ZAGREBU UNIVERSITY OF
APPLIED SCIENCES ZAGREB



**Stručni prijediplomski
studij Elektrotehnika**

3. semestar

**Komunikacijska i
računalna tehnika**

Ak.g. 2025./2026.

Signali i procesi

Fourierov red u trigonometrijskom obliku



Kalendar nastave po tjednima (ver.2)



#	Ponedjeljak	Utorak	Srijeda	Četvrtak	Petak
1	06.10.2025	07.10.2025	08.10.2025	09.10.2025	10.10.2025
2	13.10.2025	14.10.2025	15.10.2025	16.10.2025	17.10.2025
3	20.10.2025	21.10.2025	22.10.2025	23.10.2025	24.10.2025
4	27.10.2025	28.10.2025	29.10.2025	30.10.2025	31.10.2025
5	03.11.2025	04.11.2025	05.11.2025	06.11.2025	07.11.2025
6	10.11.2025	11.11.2025	12.11.2025	13.11.2025	14.11.2025
7	17.11.2025	18.11.2025	19.11.2025	20.11.2025	21.11.2025
8	24.11.2025	25.11.2025	26.11.2025	27.11.2025	28.11.2025
9	01.12.2025	02.12.2025	03.12.2025	04.12.2025	05.12.2025
10	08.12.2025	09.12.2025	10.12.2025	11.12.2025	12.12.2025
11	15.12.2025	16.12.2025	17.12.2025	18.12.2025	19.12.2025
	22.12.2025	23.12.2025	24.12.2025	25.12.2025	26.12.2025
	29.12.2025	30.12.2025	31.12.2025	01.01.2026	02.01.2026
12	05.01.2026	06.01.2026	07.01.2026	08.01.2026	09.01.2026
13	12.01.2026	13.01.2026	14.01.2026	15.01.2026	16.01.2026
14	19.01.2026	20.01.2026	21.01.2026	22.01.2026	23.01.2026
15	26.01.2026	27.01.2026	28.01.2026	29.01.2026	30.01.2026
16	02.02.2026	03.02.2026	04.02.2026	05.02.2026	06.02.2026

HRO-CIGRE

Samostalan rad

SiP 1. KZ

LV1

LV2

LV3

LV4

LV5

SiP 2. KZ

SiP LV G1
9:15 - 11:30

SiP LV G2
16:15 - 18:30

SiP PRED/AV
9:15 - 13:00

SiP LV G3
13:45 - 16:00

SiP LV G4
16:15 - 18:30

Prezentacija je samo podsjetnik o onome što je obuhvaćeno predavanjem i njezin sadržaj nije dovoljan za učenje!

Obavezna literatura – skripta:

- P. Valožić: Signali i procesi, skripta ,TVZ, 2015
Dostupno preko mojTVZ web stranice predmeta kao 8 pdf dokumenata:

Udžbenici:

- H. Hsu: Schaum's Outline of Signals and Systems
<https://uet.vnu.edu.vn/~tantd/ss/TheoryandProblemsofSS.pdf>
- H. Babić: Signali i sustavi
http://sis.zesoi.fer.hr/predavanja/pdf/sis_2001_skripta.pdf
- B. Jeren: Signali i sustavi
<https://shop.skolskaknjiga.hr/signali-i-sustavi/>
- I.S. Pandžić et al: Uvod u teoriju informacije i kodiranje (3. poglavlje)
<https://element.hr/artikli/293/uvod-u-teoriju-informacije-i-kodiranje>

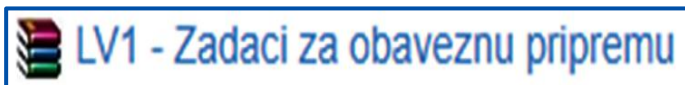
Navedena literatura je ujedno i izvor za pripremu ove prezentacije.

Z ZzV - rješenja: ZzV 07-01 ... 07-04; 08-01 ... 08-05

Laboratorijske vježbe

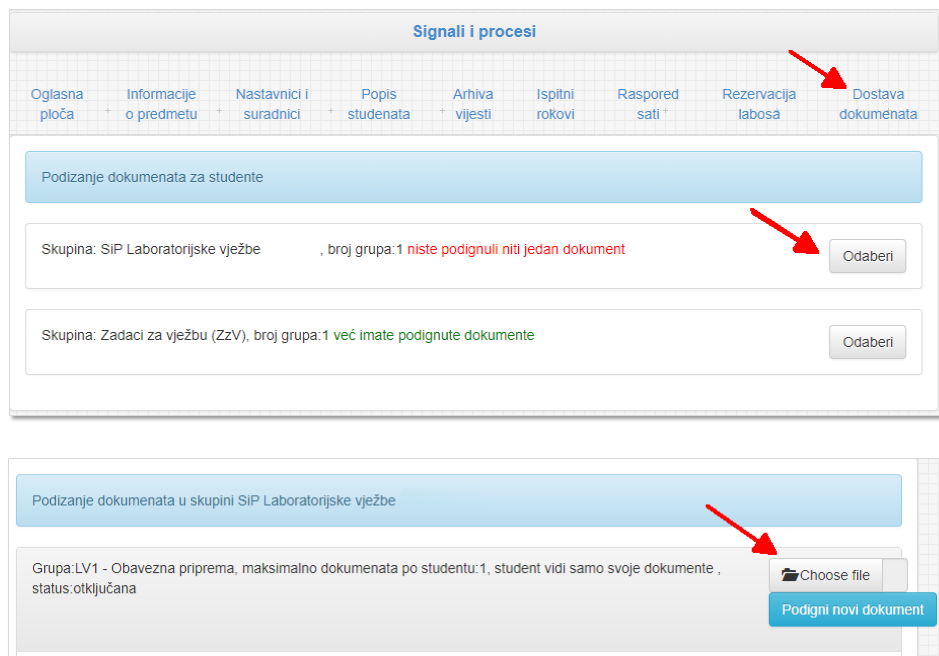
- Počinju 04.12.2025.
- Finalizirajte odabir grupa, grupe se zaključavaju 20.11.2025.
- Do početka Vašeg termina LV1 potrebno je dostaviti obaveznu pripremu
- **Bodovanje: 0 sve točno, -(2+3) netočno, -10 nije predano na vrijeme**
- Individualizirani zadaci za obaveznu pripremu bit će objavljeni u repozitoriju

Laboratorijske vježbe 2025/2026.



Laboratorijske vježbe (nastavak)

- Datoteku LV1 - Zadaci za obaveznu pripremu.zip **raspakirati i potražiti pdf dokument prema Vašem JMBAG-u**
- Ispišite pdf dokument, riješite zadatke pismeno i do početka Vašeg termina LV1 dostavite skenirano rješenje putem opcije **Dostava dokumenata na mojTVZ stranici predmeta:**



The screenshot displays the 'Signali i procesi' (Signals and Processes) section of a web application. At the top, a navigation bar includes links: Oglasna ploča, Informacije o predmetu, Nastavnici i suradnici, Popis studenata, Arhiva vijesti, Ispitni rokovi, Raspored sati, Rezervacija labosa, and Dostava dokumenata (highlighted with a red arrow). Below the navigation bar, there are two main sections for document submission. The first section, 'Podizanje dokumenata za studente' (Document upload for students), lists two groups: 'Skupina: SiP Laboratorijske vježbe' (Group: SiP Laboratory exercises) with the status 'broj grupa:1 niste podignuli niti jedan dokument' (group count: 1, you have not uploaded any documents) and an 'Odaberi' (Select) button (highlighted with a red arrow); and 'Skupina: Zadaci za vježbu (ZzV)' (Group: Exercises tasks) with the status 'broj grupa:1 već imate podignute dokumente' (group count: 1, you already have uploaded documents) and an 'Odaberi' button. The second section, 'Podizanje dokumenata u skupini SiP Laboratorijske vježbe' (Document upload in the SiP Laboratory exercises group), shows details for 'Grupa: LV1 - Obavezna priprema, maksimalno dokumenata po studentu:1, student vidi samo svoje dokumente, status:otključana' (Group: LV1 - Mandatory preparation, maximum documents per student: 1, student sees only their documents, status: unlocked) and provides two buttons: 'Choose file' and 'Podigni novi dokument' (highlighted with a red arrow).

LV1 – zadaci za obaveznu pripremu

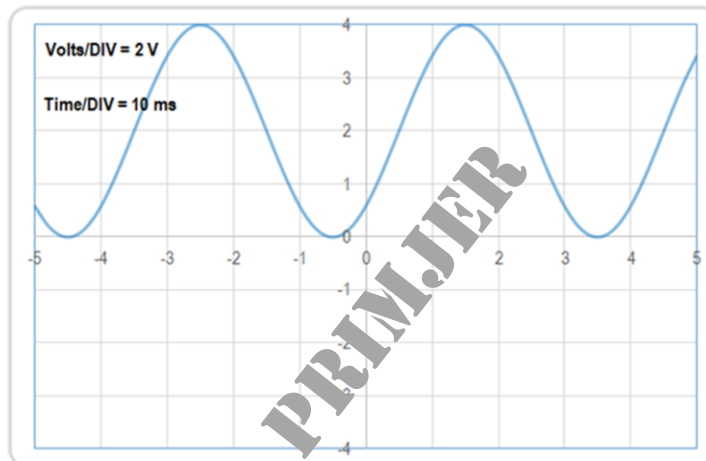
Izgled riješenog 1. zadatka



Signali i procesi - Obavezna priprema za 1. laboratorijske vježbe

Ime Prezime: Sipko Tevezeić

1. zadatak: Zapišite analitički izraz putem kosinus funkcije harmonijskog signala $x(t)$ s istosmjernom komponentom prikazanog na simuliranom ekranu osciloskopa. Fazu odredite u odnosu na središnju okomitu os i izrazite je u radijanima u intervalu $(-\pi, \pi]$.



Postupak:

Amplituda: $2 \text{ DIV} \times 2 \text{ V/DIV} = 4 \text{ V}$

DC: $2 \text{ DIV} \times 2 \text{ V/DIV} = 4 \text{ V}$

Period: $4 \text{ DIV} \times 10 \text{ ms/DIV} = 40 \text{ ms}$

Frekvencija: $1/T = 25 \text{ Hz}$

Faza: iz $x(0) = 0,6 \text{ DIV} \times 2 \text{ V/DIV} = 1,2 \text{ V}$

$$4\cos(\varphi) + 4 = 1,2$$

$$\varphi = \arccos(-0,7) = 2,346$$

Ali faza je negativna (zašto!?) $\Rightarrow \varphi = -2,346 \text{ rad}$

kao razlomak: $\varphi = -3/4 \pi \text{ rad}$

ili na neki drugi način...

$$A [\text{V}] = 4$$

$$DC [\text{V}] = 4$$

$$f [\text{Hz}] = 25$$

$$\varphi [\text{rad}] = -2,35$$

$$\varphi [p/q \pi \text{ rad}] = -3/4 \pi$$

Rješenje:

$$x(t) = 4\cos(2\pi \cdot 25t - 3/4\pi) + 4 \text{ [V]}$$

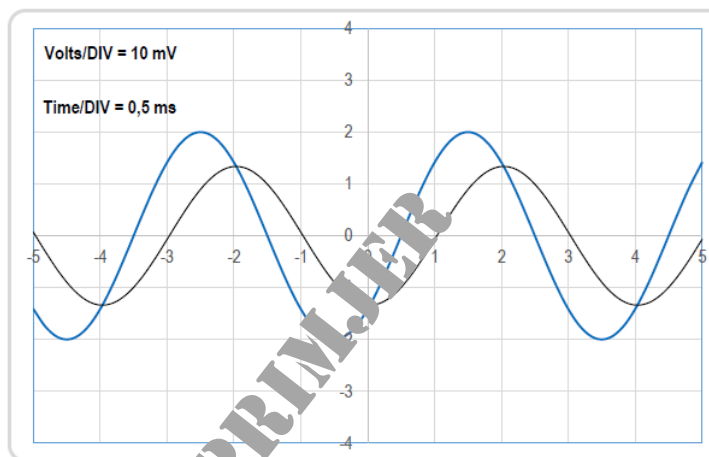
LV1 – zadaci za obaveznu pripremu

Izgled 2. zadatka



2. zadatak:

Odredite $H_a = A_{iz} / A_{ul}$ i $H_p = \varphi_{iz} - \varphi_{ul}$ očitavanjem vrijednosti na simuliranom ekranu osciloskopa (signal na ulazu u filter prikazan je plavom bojom, a na izlazu crnom bojom). O kojem tipu filtera je riječ? Na temelju dobivenih vrijednosti za H_a i H_p izračunajte uz pomoć odgovarajućih formula graničnu frekvenciju filtera f_0 i vrijednost kondenzatora C , ako je $R = 1 \text{ k}\Omega$.



Postupak:

H_a [V/V] = _____

H_a [dB] = _____

H_p [rad] = _____

Tip filtera = _____

Frekvencija signala f [Hz] = _____

f_0 iz vrijednosti H_a [Hz] = _____

f_0 iz vrijednosti H_p [Hz] = _____

C iz prosj. vrijednosti za f_0 i R [nF] = _____

Datum: _____

Potpis: Sipko Tevezić

Uvod u Fourierov red u trigonometrijskom obliku

Analiza i sinteza signala

Složen je onaj signal kojega je moguće **analizirati**, rastaviti na jednostavne (elementarne) signale ili je pak je nastao **sintezom** – kombiniranjem jednostavnih (elementarnih) signala.

Odgovor na pitanje: Je li određen signal jednostavan ili složen? – relativan je i ovisi o odabranoj skupini elementarnih signala, tzv. **bazi**.

Različiti mogu biti i načini kako složeni signal sastaviti – sintetizirati od jednostavnih. Postupke sinteze možemo podijeliti u dvije skupine: **linearne** i **nelinearne**.

Od linearnih postupaka sinteze u nastavku opisana je **linearna kombinacija**, a od nelinearnih – **modulacija**.

Linearna kombinacija funkcija: **Kontinuirani signali**

$$y(t) = a_0 y_0(t) + a_1 y_1(t) + \dots + a_n y_n(t)$$

$$a_k = \frac{\int_{t_1}^{t_2} y(t) \cdot y_k(t) \cdot dt}{\int_{t_1}^{t_2} y_k^2(t) \cdot dt}$$

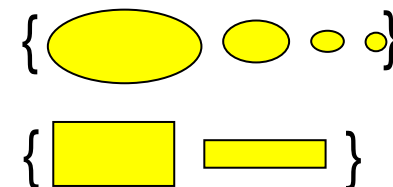
Linearna kombinacija sekvenci: **Digitalni signali**

$$y_i = a_0 y_{0i} + a_1 y_{1i} + \dots + a_n y_{ni}$$

$$a_k = \frac{\sum_{i_1}^{i_2} y_i \cdot y_{ki}}{\sum_{i_1}^{i_2} y_{ki}^2}$$

Prikaz složenog signala $y(t)$ kao **linearna kombinacija** jednostavnih, **elementarnih** signala:

$$y(t) = a_0 y_0(t) + a_1 y_1(t) + \dots + a_n y_n(t)$$



Skup elementarnih signala $\{y_i(t)\}$ tvori **bazu**. Neovisna varijabla najčešće je vrijeme t . Čest izuzetak su slike – dvodimenzionalne poruke s bazama signala kojima su neovisne varijable prostorne koordinate: širina (x) ili visina (y).

Interpretacija: Signal $y(t)$ dobiti ćemo ako uzmemo a_0 signala $y_0(t)$, tome dodamo a_1 signala $y_1(t)$, do elementarnog signala $y_n(t)$ kojega dodamo prethodnom zbroju u količini a_n . Nalik na kemijsku smjesu - jedini postupak je zbrajanje komponenti. Koeficijente a_i možemo proglasiti "težinskim udjelima" elementarnih signala u "smjesi".

Sinteza

U postupku sinteze moraju biti poznati sastojci (baza $\{y_i(t)\}$) i težinski koeficijenti $\{a_0, a_1, \dots, a_n\}$, pa će rezultat biti jedinstven signal $y(t)$.

$$y(t) \Leftarrow a_0 y_0(t) + a_1 y_1(t) + \dots + a_n y_n(t)$$

Analiza

Obratni postupak, kada uz poznatu bazu $\{y_i(t)\}$, za zadani signal $y(t)$ želimo odrediti (izračunati ili izmjeriti) težinske udjele $\{a_0, a_1, \dots, a_n\}$, pronaći koliko (a_i) kojeg elementarnog signala $y_i(t)$ ima u linearnoj kombinaciji („smjesi”) $y(t)$.

Kod zbrajanja elementarnih signala, donja granica cjelobrojnog indeksa i može biti $-\infty$, $-n$ ili 0 , dok gornja granica može biti n ili $+\infty$. Tada imamo prebrojivo mnogo (konačno ili beskonačno) "sastojaka" složenog signala, $\Rightarrow$ skup elementarnih signala je **diskretan**.

Sastavnica (komponenata) $a_i \cdot y_i$ može biti neprebrojivo mnogo. Tada suma u izrazu za linearnu kombinaciju postaje **integral**, npr.:

$$y(t) = \int_{-\infty}^{\infty} a(f) \cdot y(f, t) df$$

Težinski koeficijenti a_i više nisu diskretni i nije ih moguće numerirati indeksom i , te postaju težinska funkcija drugog kontinuiranog argumenta npr. frekvencije $a(f)$ ili kružne frekvencije $a(\omega)$. Jednostavni signali y funkcija su dvije varijable, npr. t i f (ili t i ω).

Analiza složenih signala lagana je zadaća kada elementarni signali tvore **ortonormiranu** bazu. Pojedinačni elementarni signali $x(t)$ i $y(t)$ uzajamno su ortogonalni, ako je njihov skalarni produkt jednak nuli:

$$(x(t), y(t)) = \int_t x(t) y(t) dt = 0$$

$$(X, Y) = \sum_i X_i \cdot Y_i = 0$$

Primjer dvaju ortogonalnih signala jest $x(t)=\sin(\omega \cdot t)$ i $y(t)=\cos(\omega \cdot t)$.
Uvrštavanjem u izraz za skalarani produkt, dobiva se:

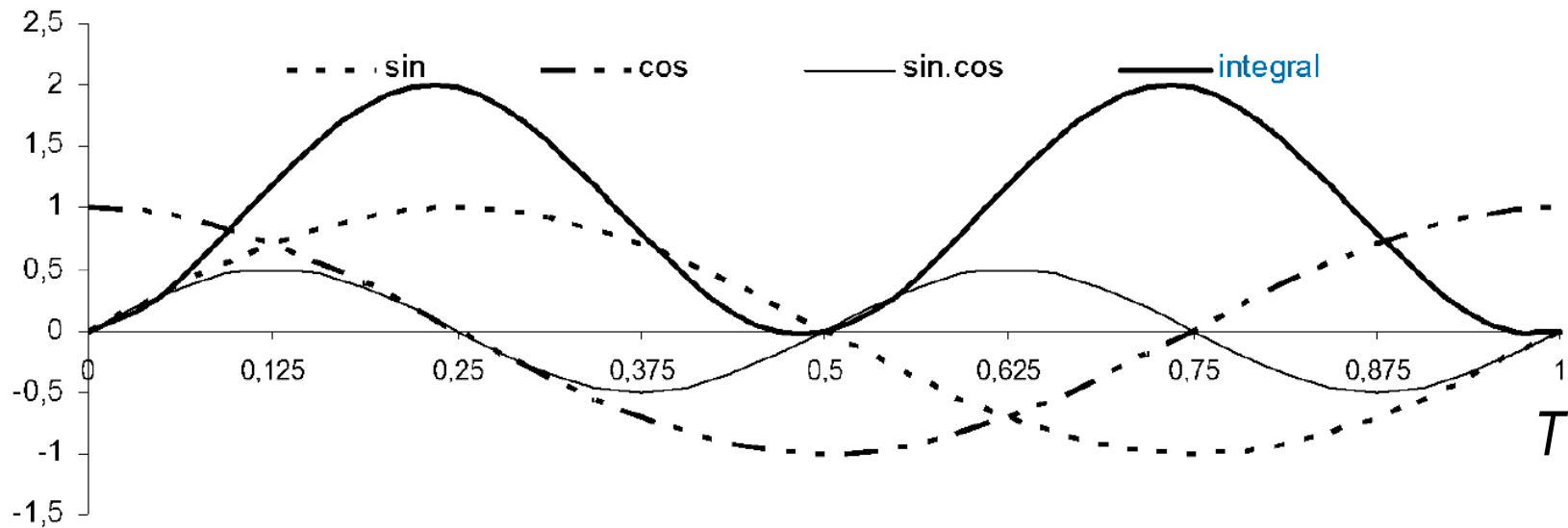
$$(x(t), y(t)) = \int_T \sin(\omega \cdot t) \cos(\omega \cdot t) dt$$

$$(x(t), y(t)) = \frac{1}{2} \int_T \sin(2 \cdot \omega \cdot t) dt$$

$$(x(t), y(t)) = -\frac{1}{2} \frac{1}{2 \cdot \omega} \cos(2 \cdot \omega \cdot t) \Big|_0^T$$

$$(x(t), y(t)) = 0$$

Dakle, signali $x(t)=\sin(\omega \cdot t)$ i $y(t)=\cos(\omega \cdot t)$ ortogonalni su na periodu T , ali i na polovici period $T/2$, što je vidljivo iz grafa u nastavku (krivulja s punom ispunom označena s [integral](#))



Kao ortogonalne baze za analizu signala koristit ćemo harmonijske funkcije ($\sin$ i/ili $\cos$) u trigonometrijskom ili eksponencijalnom obliku, te Diracove δ - funkcije.

Normirana baza

U slučaju kad skalarni produkt elementarnog vektora normirane baze, samog sa sobom daje vrijednost kvadrata modula (norme) signala jednak 1, bazu nazivamo **normirana baza**:

$$|x(t)|^2 = (x(t), x(t)) = \int_t x(t) x(t) dt = 1$$
$$|x|^2 = (x, x) = \sum_i x_i x_i = 1$$

Ortonormirana baza

U slučaju kad je baza ortogonalna i normirana, nazivamo je **ortonormirana**.

Fourierov red u trigonometrijskom obliku

Fourierova analiza kao bazu ima **ortogonalan** skup harmonijskih (sinusnih i kosinusnih) signala $y_i(t)$.

Rastaviti (analizirati) složeni signal, znači odrediti (izračunati ili izmjeriti) koeficijente a_i :

$$y(t) = \sum_i a_i y_i(t)$$

$$a_k = \frac{\int_{t_1}^{t_2} y(t) \cdot y_k(t) \cdot dt}{\int_{t_1}^{t_2} y_k^2(t) \cdot dt}$$

Ako je složeni signal **periodičan s periodom T** , tada su:

- Sve komponente periodične sa T ;
- Period komponenti može biti T ili T/n , gdje je $n \in \mathbf{N}$;
- Frekvencije jednostavnih (harmonijskih) signala, cjelobrojni su umnožak najniže tzv. **osnovne** frekvencije f_1 :

$$f_1 = 1/T, \quad \omega_1 = 2 \cdot \pi \cdot f_1 = 2 \cdot \pi / T$$

Fourierov red u trigonometrijskom obliku



Koeficijenti a_i i b_i predstavljaju amplitude kosinusnih i sinusnih članova reda, a računaju se primjenom sljedećih izraza:

$$a_i = \frac{2}{T} \int_T x(t) \cos(i \cdot \omega_1 t) dt$$

$$i = 1, 2, 3, \dots$$

$$b_i = \frac{2}{T} \int_T x(t) \sin(i \cdot \omega_1 t) dt$$



Iz jednadžbe za a_i uvrstivši za $i = 0$, dobijemo:

$$a_0 = \frac{2}{T} \int_T x(t) dt$$

$$\int_{t_0}^{t_0+T} f(t) dt = \int_T f(t) dt$$

$$a_k = \frac{\int_{t_1}^{t_2} y(t) \cdot y_k(t) \cdot dt}{\underbrace{\int_{t_1}^{t_2} y_k^2(t) \cdot dt}}$$

$$\int_0^T \cos^2\left(\frac{2\pi x}{T}\right) dx = \frac{T}{2}$$

$$\int_0^T \sin^2\left(\frac{2\pi x}{T}\right) dx = \frac{T}{2}$$

Fourierov red u **trigonometrijskom** obliku:

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} [a_i \cos(i \cdot \omega_1 t) + b_i \sin(i \cdot \omega_1 t)] \quad i = 1, 2, 3, \dots$$



$$x(t) = c_0 + \sum_{i=1}^{+\infty} c_i \cos(i \cdot \omega_1 t + \varphi_i) \quad \left\{ \begin{array}{l} c_0 = a_0/2 \\ c_i = \sqrt{a_i^2 + b_i^2} ; i > 0 \\ \varphi_i = -\text{atan2}(a_i, b_i) = \text{atan2}(a_i, -b_i) \end{array} \right.$$

$\varphi_i = -\arctg(b_i/a_i)$ za 1. i 4. kvadrant

Fourierovim redom, složeni signal **x(t)** prikazan je kao linearna kombinacija:

- istosmjerne komponente (veliĉine $a_0/2$ i frekvencije $\omega_0=0$)
- osnovnih signala: "cos" i "sin" signala frekvencije ω_1 ($i = 1$)
- viših harmonika frekvencije $2 \cdot \omega_1, 3 \cdot \omega_1, 4 \cdot \omega_1, \dots i \cdot \omega_1 \dots$ ($i = 2, 3, \dots$)

$$\varphi_i = \text{atan2}(a_i, -b_i)$$

Izračun **atan2** s pomoću **arctg**:

Za I. kvadrant ($a_i > 0$ i $b_i \geq 0$) $\Rightarrow \varphi_i = -\text{arctg}(b_i/a_i)$

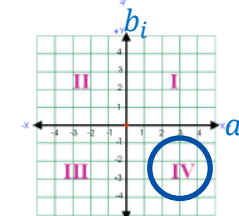
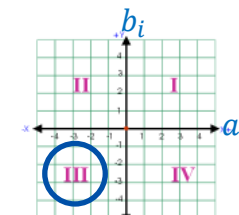
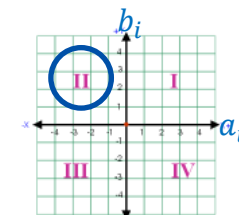
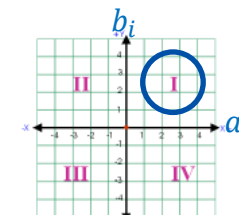
za $a_i = 0$ i $b_i > 0 \Rightarrow \varphi_i = -\pi/2$

Za II. kvadrant ($a_i < 0$ i $b_i \geq 0$) $\Rightarrow \varphi_i = -\text{arctg}(b_i/a_i) - \pi$

Za III. kvadrant ($a_i < 0$ i $b_i \leq 0$) $\Rightarrow \varphi_i = -\text{arctg}(b_i/a_i) + \pi$

za $a_i = 0$ i $b_i > 0 \Rightarrow \varphi_i = +\pi/2$

Za IV. kvadrant ($a_i > 0$ i $b_i \leq 0$) $\Rightarrow \varphi_i = -\text{arctg}(b_i/a_i)$



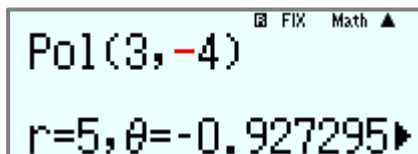
Fourierov red u trigonometrijskom obliku



Izračun φ_i iz a_i i b_i : $\varphi_i = \text{atan2}(a_i, -b_i)$; Kalkulatorom $\Rightarrow$ **Pol** (a_i , $-b_i$)

$$a_i = +3, b_i = +4$$

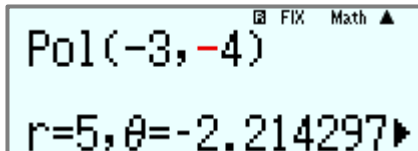
$$\varphi = \text{atan2}(+3, -4) = -0,9273$$



Pol(3, -4)
r=5, $\theta = -0.927295$

$$a_i = -3, b_i = +4$$

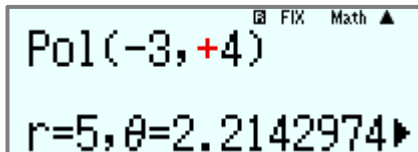
$$\varphi = \text{atan2}(-3, -4) = -2,2143$$



Pol(-3, -4)
r=5, $\theta = -2.214297$

$$a_i = -3, b_i = -4$$

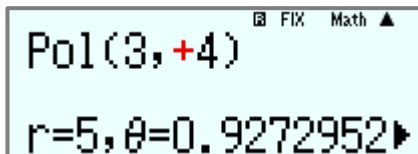
$$\varphi = \text{atan2}(-3, +4) = -2,2143$$



Pol(-3, +4)
r=5, $\theta = 2.2142974$

$$a_i = +3, b_i = -4$$

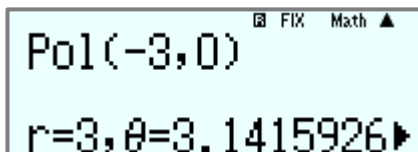
$$\varphi = \text{atan2}(+3, +4) = +0,9273$$



Pol(3, +4)
r=5, $\theta = 0.9272952$

$$a_i = -3, b_i = 0$$

$$\varphi = \text{atan2}(-3, 0) = +3,14159$$



Pol(-3, 0)
r=3, $\theta = 3.1415926$

Fourierov red u **trigonometrijskom** obliku:

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} [a_i \cos(i \cdot \omega_1 t) + b_i \sin(i \cdot \omega_1 t)] \quad i = 1, 2, 3, \dots$$

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} c_i \sin(i \cdot \omega_1 t + \psi_i) \quad \left\{ \begin{array}{l} c_i = \sqrt{a_i^2 + b_i^2} \\ \psi_i = +\text{atan2}(b_i, a_i) \end{array} \right.$$

$\psi_i = \arctg(a_i/b_i)$ za 1. i 2. kvadrant

Fourierovim redom, složeni signal **x(t)** prikazan je kao linearna kombinacija:

- istosmjerne komponente (veliĉine $a_0/2$ i frekvencije $\omega_0=0$)
- osnovnih signala: "cos" i "sin" signala frekvencije ω_1 ($i = 1$)
- viših harmonika frekvencije $2 \cdot \omega_1, 3 \cdot \omega_1, 4 \cdot \omega_1, \dots i \cdot \omega_1 \dots$ ($i = 2, 3, \dots$)

Za one koji žele znati više: Fourierov r. preko sinusa



$$\psi_i = +\text{atan2}(b_i, a_i)$$

Izračun **atan2** s pomoću **arctg**:

Za I. kvadrant ($a_i > 0$ i $b_i \geq 0$) $\Rightarrow \psi_i = \text{arctg}(a_i/b_i)$

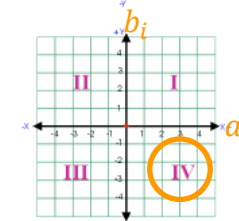
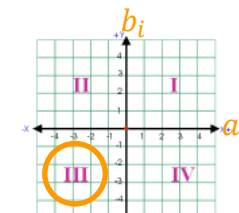
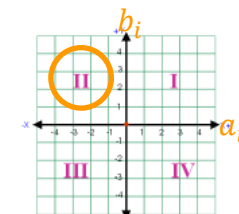
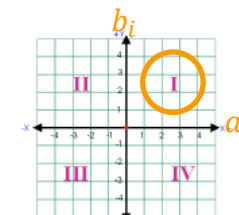
za $a_i > 0$ i $b_i = 0 \Rightarrow \psi_i = +\pi/2$

Za II. kvadrant ($a_i \leq 0$ i $b_i > 0$) $\Rightarrow \psi_i = \text{arctg}(a_i/b_i)$

za $a_i < 0$ i $b_i = 0 \Rightarrow \psi_i = -\pi/2$

Za III. kvadrant ($a_i \leq 0$ i $b_i < 0$) $\Rightarrow \psi_i = \text{arctg}(a_i/b_i) - \pi$

Za IV. kvadrant ($a_i > 0$ i $b_i < 0$) $\Rightarrow \psi_i = \text{arctg}(a_i/b_i) + \pi$



Za one koji žele znati više: Fourierov r. preko sinusa



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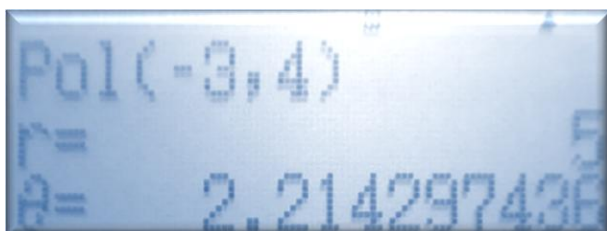
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Izračun $\psi_i = +\text{atan2}(b_i, a_i)$ kalkulatorom $\Rightarrow$ **Pol** (b_i, a_i):

$$\psi = \text{atan2}(+3, +4) \\ = +0,9273$$



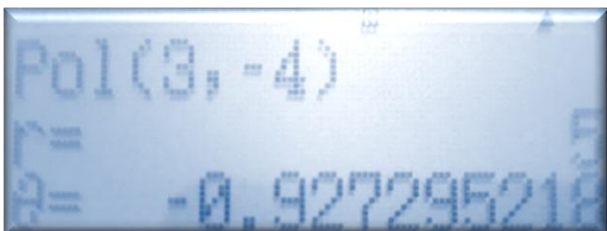
$$\psi = \text{atan2}(-3, +4) \\ = 2,2143$$



$$\psi = \text{atan2}(-3, -4) \\ = -2,2143$$

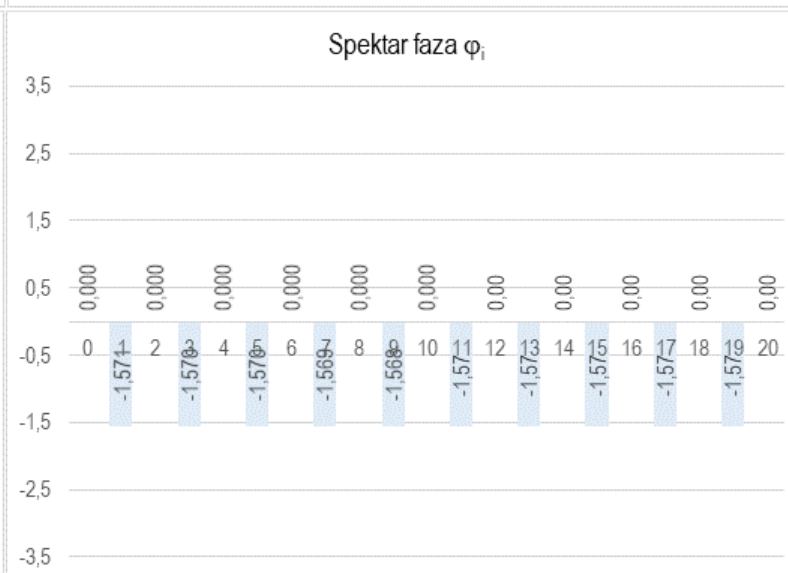
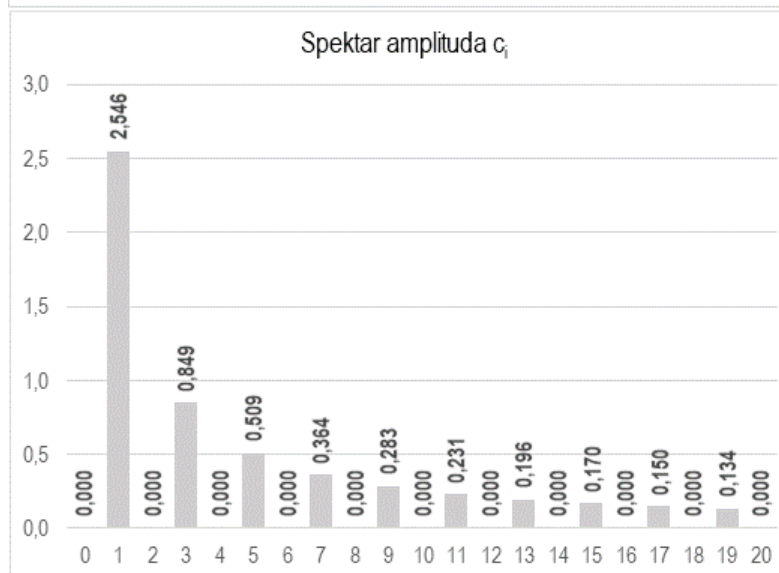
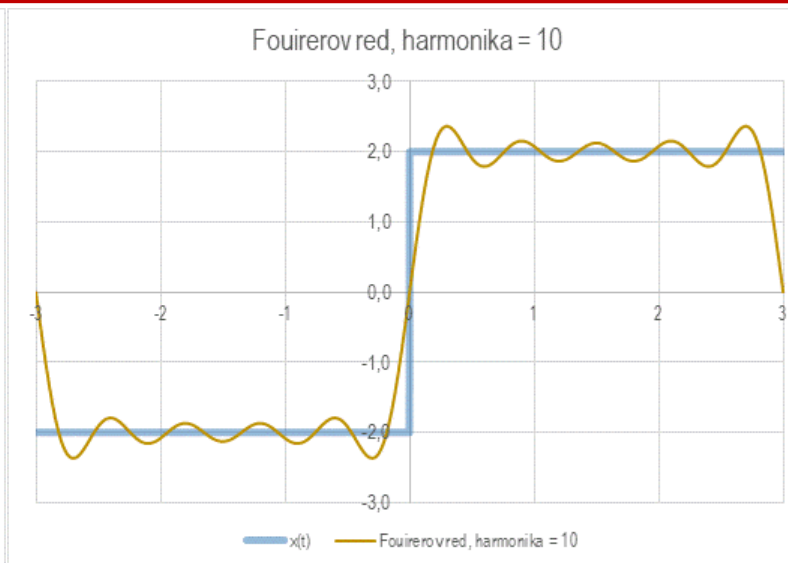
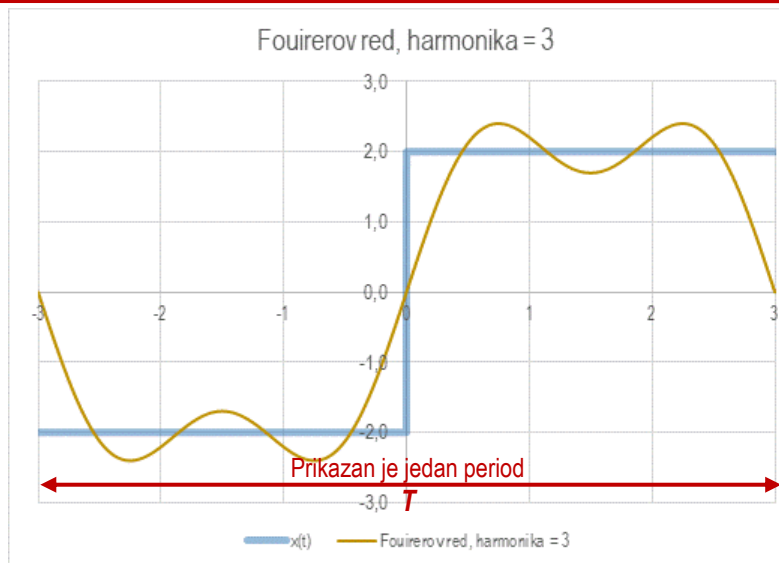


$$\psi = \text{atan2}(+3, -4) \\ = -0,9273$$



Primjer 1. Neparna funkcija

Primjer 1. Neparna funkcija



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 1.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
0,000	0,000	2,546	0,000	0,000	0,000	0,849	0,000	0,000	0,000	0,509	0,000	0,000	0,000	0,364	0,000	0,000	0,000	0,283	0,000	0,000

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	0,000	2,546	0,000	0,849	0,000	0,509	0,000	0,364	0,000	0,283	0,000
$f_{i_j} =$	0,000	-1,571	0,000	-1,570	0,000	-1,570	0,000	-1,569	0,000	-1,568	0,000

Parsevalov teorem:

Zašto su $a_i = 0$?

Zašto nema parnih harmonika?
(pogledati sljedeću stranicu!)

Zašto je faza 0 ili $-\pi/2$?

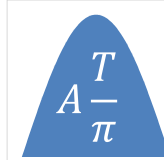
$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 3,8384$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 4$$

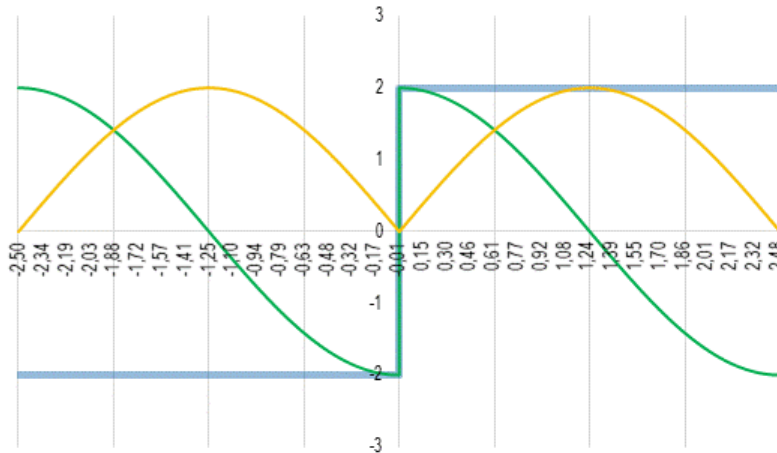
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 1.



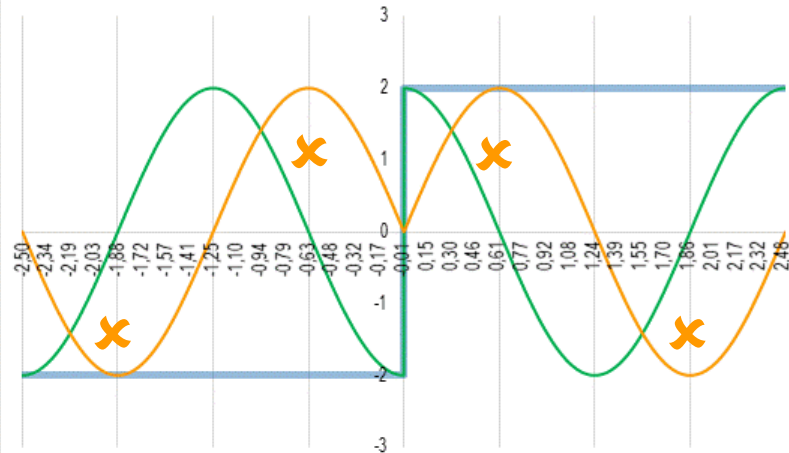
$$(2/T) \cdot \int x(t) \cdot \cos(1 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(1 \cdot \omega t) dt = 2,55$$



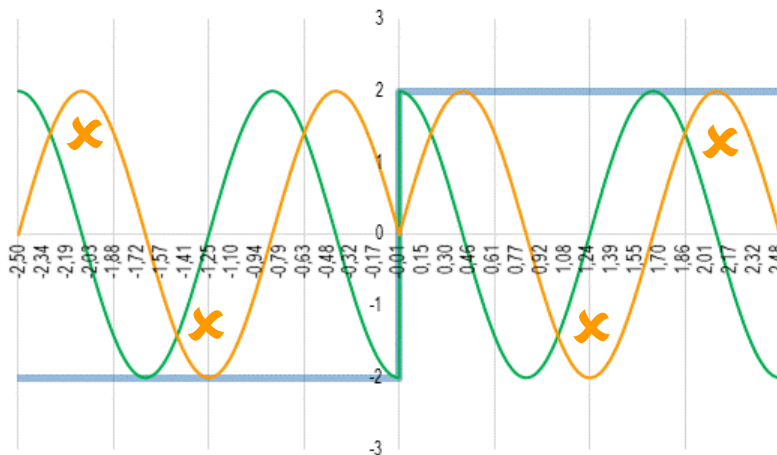
$$(2/T) \cdot \int x(t) \cdot \cos(2 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(2 \cdot \omega t) dt = 0$$



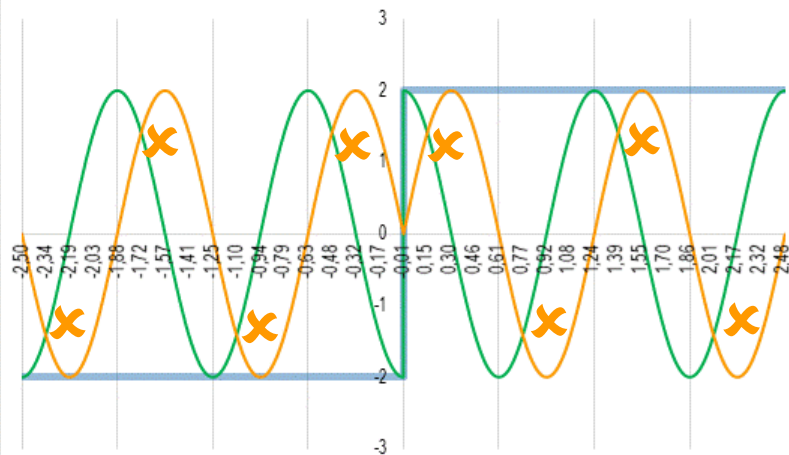
$$(2/T) \cdot \int x(t) \cdot \cos(3 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(3 \cdot \omega t) dt = 0,85$$



$$(2/T) \cdot \int x(t) \cdot \cos(4 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(4 \cdot \omega t) dt = 0$$



Primjer 1. Grafičko integriranje



Integral poluperioda sinusnog signala iznosi:

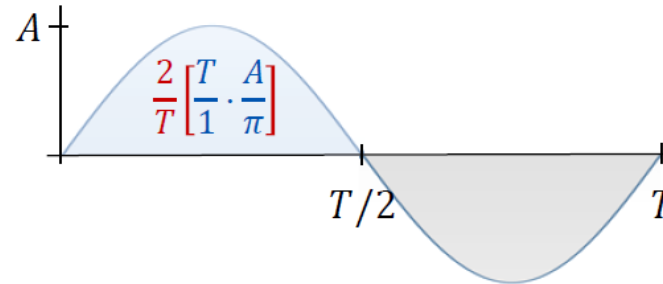
$$\int_0^{T/2} A \sin\left(\frac{2\pi}{T}t\right) dt = T \frac{A}{\pi}$$

Što se može se iskoristiti za **brzo** „grafičko integriranje” harmonijskog signala prilikom izračuna koeficijenata Fourierovog reda za **pravokutni signal**:

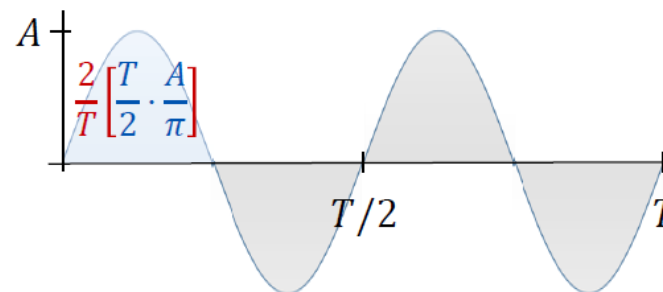
1. Unutar intervala perioda T , međusobno poništiti pozitivne i negativne brjegove
2. Koeficijent Fourierovog reda, a_n ili b_n je broj „preostalih” brjegova pomnožen s: $\frac{2A}{n\pi}$

Koliko iznosi $\int_0^{\pi} \sin t \, dt$?

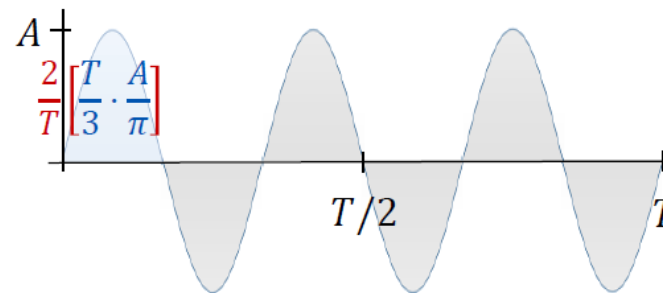
1. harmonik



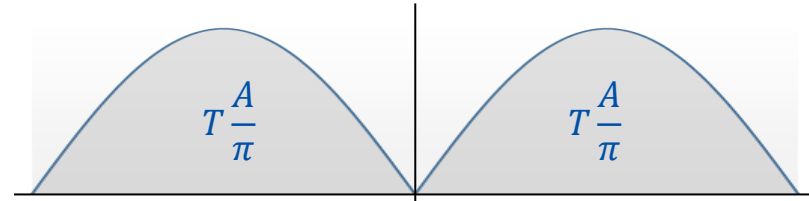
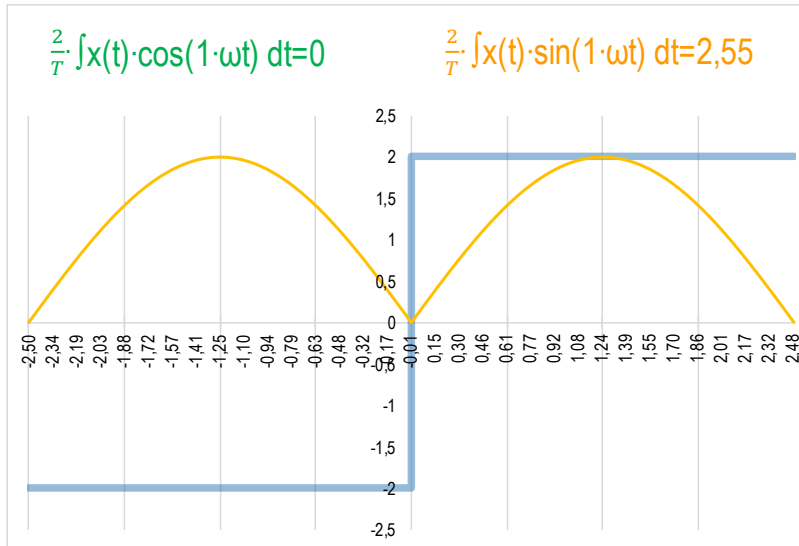
2. harmonik



3. harmonik

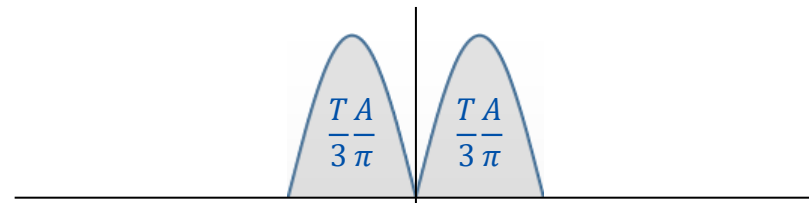
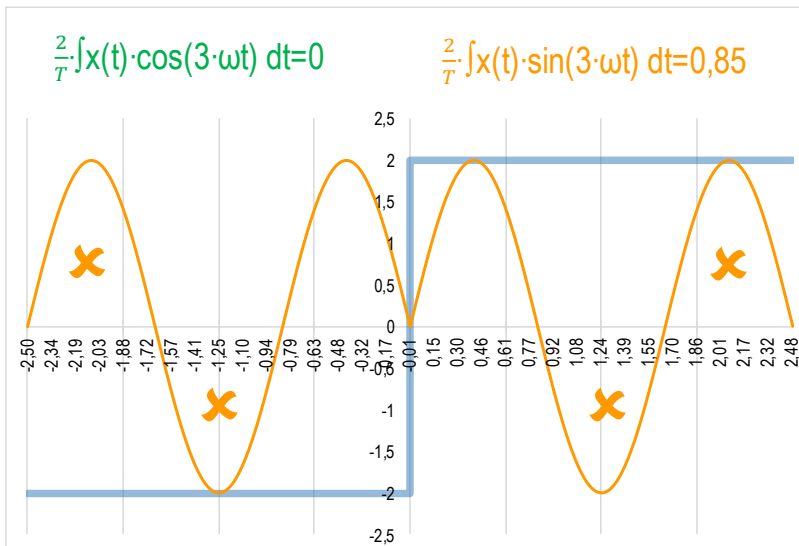


Primjer 1. Grafičko integriranje



$$b_1 = \frac{2}{T} \left[2 \cdot A \frac{T}{\pi} \right] = \frac{8}{\pi} = 2,546$$

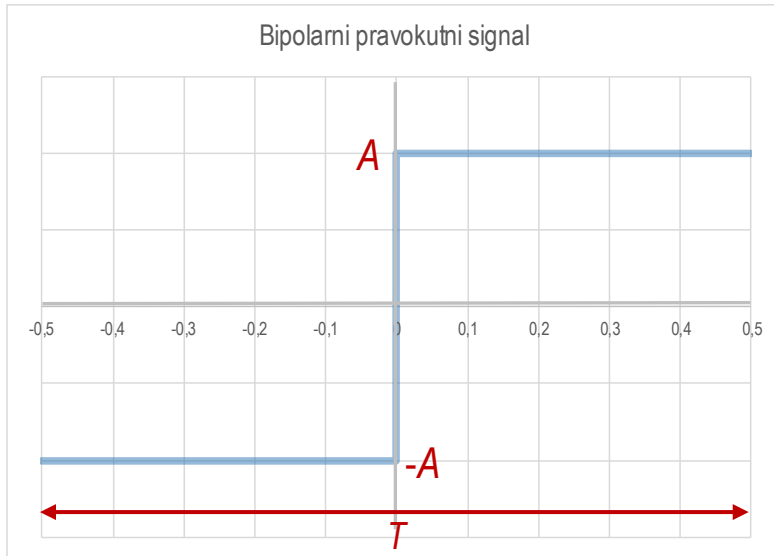
$$b_1 = 2 \frac{2A}{1 \cdot \pi} = \frac{8}{\pi} = 2,546$$



$$b_3 = \frac{2}{T} \left[2 \cdot \frac{T A}{3 \pi} \right] 2 \cdot \frac{2}{T} A \frac{T}{3 \pi} = \frac{8}{3 \pi} = 0,849$$

$$b_3 = 2 \frac{2A}{3 \cdot \pi} = \frac{8}{3 \pi} = 0,849$$

Primjer 1. Analitičko rješenje



$$a_0 = \frac{2}{T} \int_T x(t) dt = 0$$

$$a_i = \frac{2}{T} \int_T x(t) \cos(i \cdot \omega_1 t) dt$$

$$i = 1, 2, 3, \dots$$

$$b_i = \frac{2}{T} \int_T x(t) \sin(i \cdot \omega_1 t) dt$$

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} [a_i \cos(i \cdot \omega_1 t) + b_i \sin(i \cdot \omega_1 t)]$$

$$x(t) = c_0 + \sum_{i=1}^{+\infty} c_i \cos(i \cdot \omega_1 t + \varphi_i) \quad \left\{ \begin{array}{l} c_i = \sqrt{a_i^2 + b_i^2} ; i > 0 \\ \varphi_i = -\text{atan2}(a_i, b_i) \end{array} \right.$$

Primjer 1. Analitičko rješenje



$$a_0 = \frac{2}{T} \int_T x(t) dt = 0$$

$$\text{Podsjetnik: } \int \sin(ax) dx = -\frac{\cos(ax)}{a} + C \quad \int \cos(ax) dx = \frac{\sin(ax)}{a} + C$$

$$a_i = \frac{2}{T} \int_T x(t) \cos(i \cdot \omega_1 t) dt = 0 \quad \text{Neparna funkcija } x(t)! \quad i = 1, 2, 3, \dots$$

$$b_i = \frac{2}{T} \int_T x(t) \sin(i \cdot \omega_1 t) dt = \frac{2}{T} \left[\int_{-T/2}^0 -A \sin(i \cdot \omega_1 t) dt + \int_0^{T/2} A \sin(i \cdot \omega_1 t) dt \right]$$

$$b_i = \frac{2}{T} \left[(-A) \left(-\frac{\cos\left(i \cdot \frac{2\pi}{T} t\right)}{i \cdot \frac{2\pi}{T}} \right) \Big|_{-T/2}^0 + A \left(-\frac{\cos\left(i \cdot \frac{2\pi}{T} t\right)}{i \cdot \frac{2\pi}{T}} \right) \Big|_0^{T/2} \right]$$

Primjer 1. Analitičko rješenje

$$b_i = \frac{2}{T} \left[(-A) \left(-\frac{\cos\left(i \cdot \frac{2\pi}{T} t\right)}{i \cdot \frac{2\pi}{T}} \right) \Big|_{-T/2}^0 + A \left(-\frac{\cos\left(i \cdot \frac{2\pi}{T} t\right)}{i \cdot \frac{2\pi}{T}} \right) \Big|_0^{T/2} \right]$$

$$b_i = \frac{A}{i \cdot \pi} \left[\cos\left(i \cdot \frac{2\pi}{T} t\right) \Big|_{-T/2}^0 - \cos\left(i \cdot \frac{2\pi}{T} t\right) \Big|_0^{T/2} \right]$$

$$b_i = \frac{A}{i \cdot \pi} \left[\cos(0) - \cos\left(i \cdot \frac{2\pi}{T} \left(-\frac{T}{2}\right)\right) - \cos\left(i \cdot \frac{2\pi}{T} \left(\frac{T}{2}\right)\right) + \cos(0) \right]$$

$$b_i = \frac{2A}{i \cdot \pi} [1 - \cos(i\pi)]$$

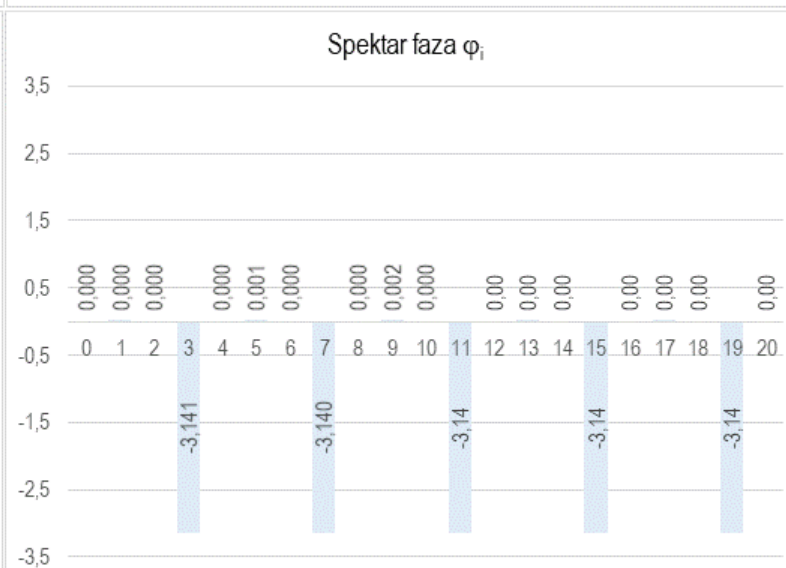
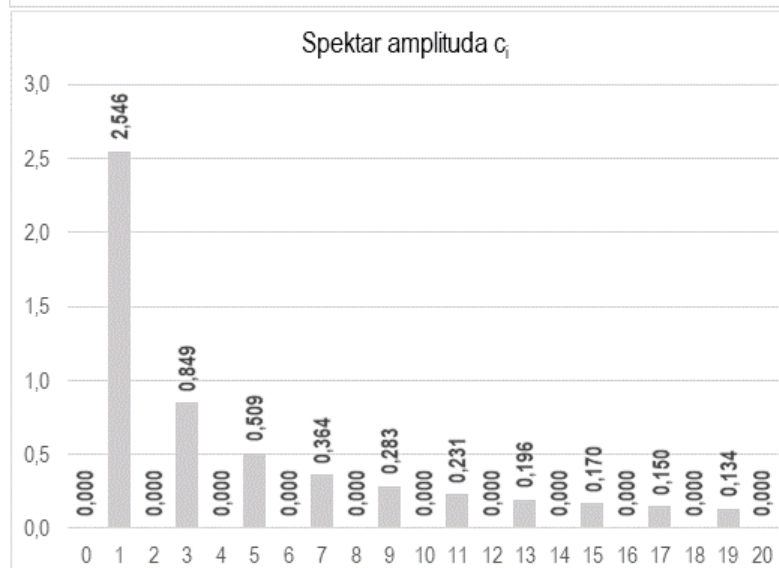
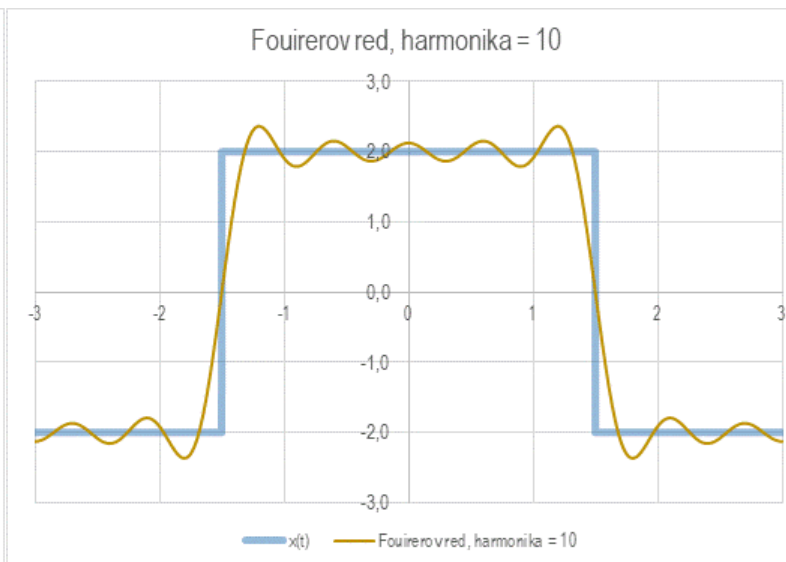
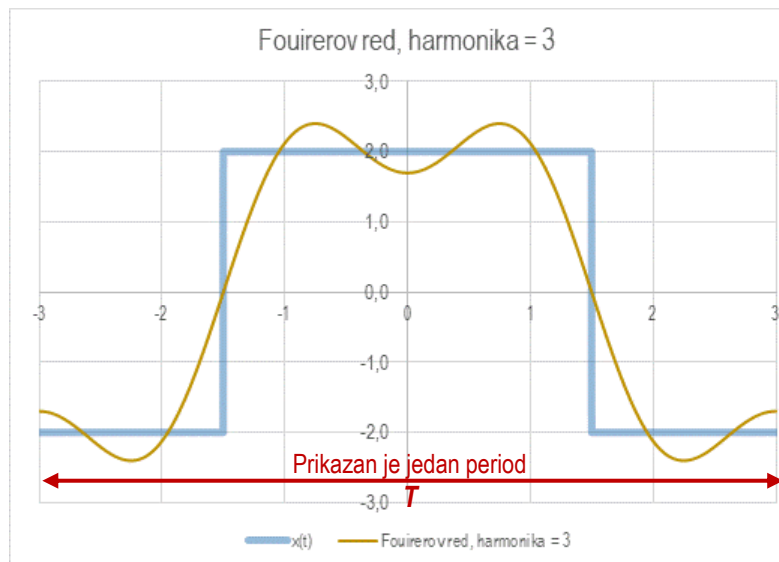
$$\left. \begin{aligned} b_i &= \frac{4A}{i \cdot \pi}, \text{ za } i = 1, 3, 5, \dots \\ b_i &= 0, \text{ za } i = 2, 4, 6, \dots \end{aligned} \right\}$$

$$c_i = \frac{4A}{i \cdot \pi} \quad \varphi_i = -\pi/2 \quad \text{za } i = 1, 3, 5, \dots$$

$$c_i = 0 \quad \varphi_i = 0 \quad \text{za } i = 0, 2, 4, \dots$$

Primjer 2. Parna funkcija

Primjer 2. Parna funkcija



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 2.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
0,000	2,546	0,000	0,000	0,000	-0,849	0,000	0,000	0,000	0,509	0,000	0,000	0,000	-0,364	0,000	0,000	0,000	0,283	0,000	0,000	0,000

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	0,000	2,546	0,000	0,849	0,000	0,509	0,000	0,364	0,000	0,283	0,000
$f_{i_j} =$	0,000	0,000	0,000	-3,141	0,000	0,001	0,000	-3,140	0,000	0,002	0,000

Zašto su $b_i = 0$?

Zašto nema parnih harmonika?

Zašto je faza $\varphi_1 = 0$?

Zašto je faza 0 ili $-\pi$?

Bi li mogla biti faza π umjesto izračunatog $-\pi$?

Razlikuje li se spektar amplituda od Primjera 1.?

Parsevalov teorem:

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 3,8384$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 4$$

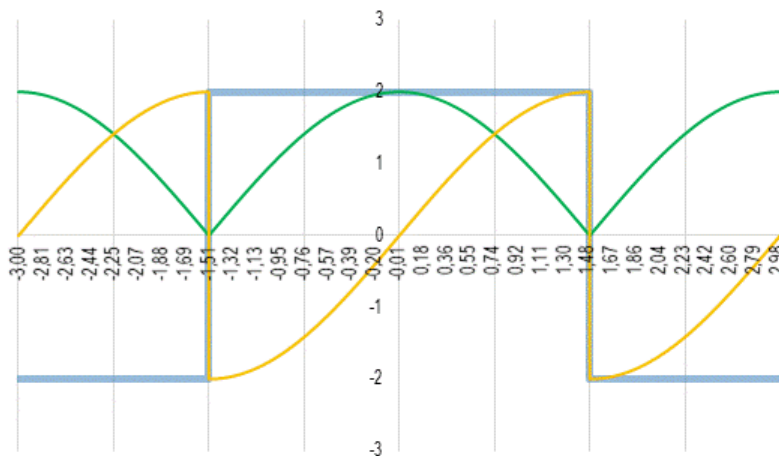
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 2.



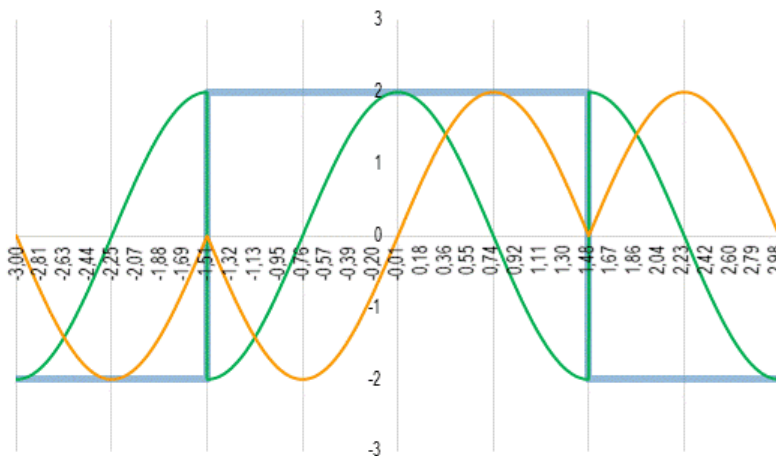
$$(2/T) \cdot \int x(t) \cdot \cos(1 \cdot \omega t) dt = 2,55$$

$$(2/T) \cdot \int x(t) \cdot \sin(1 \cdot \omega t) dt = 0$$



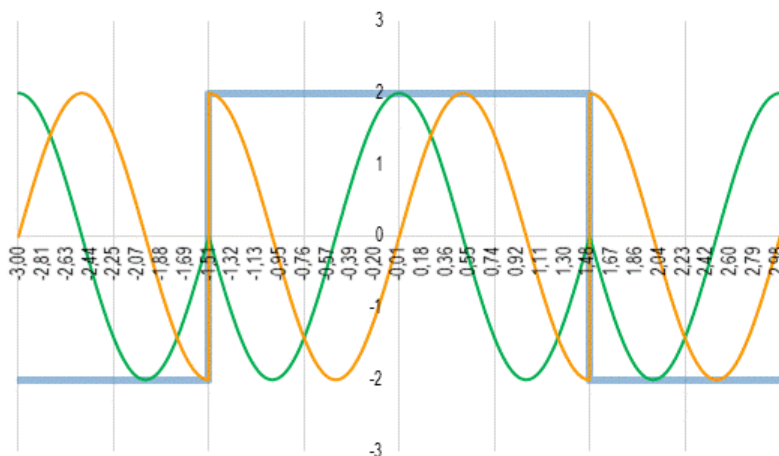
$$(2/T) \cdot \int x(t) \cdot \cos(2 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(2 \cdot \omega t) dt = 0$$



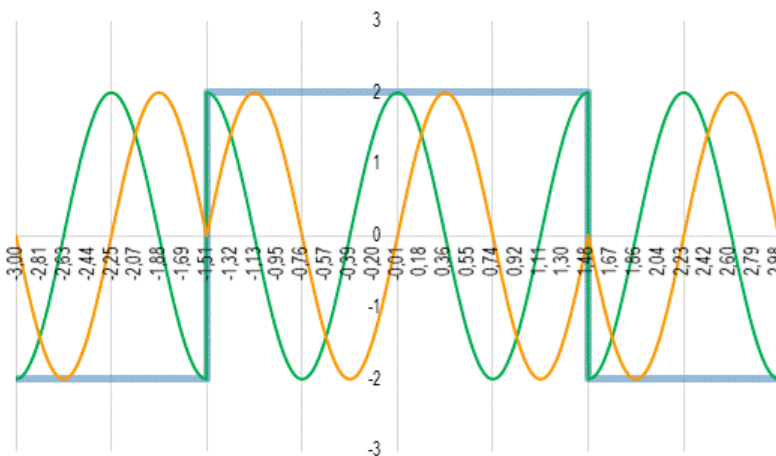
$$(2/T) \cdot \int x(t) \cdot \cos(3 \cdot \omega t) dt = -0,85$$

$$(2/T) \cdot \int x(t) \cdot \sin(3 \cdot \omega t) dt = 0$$



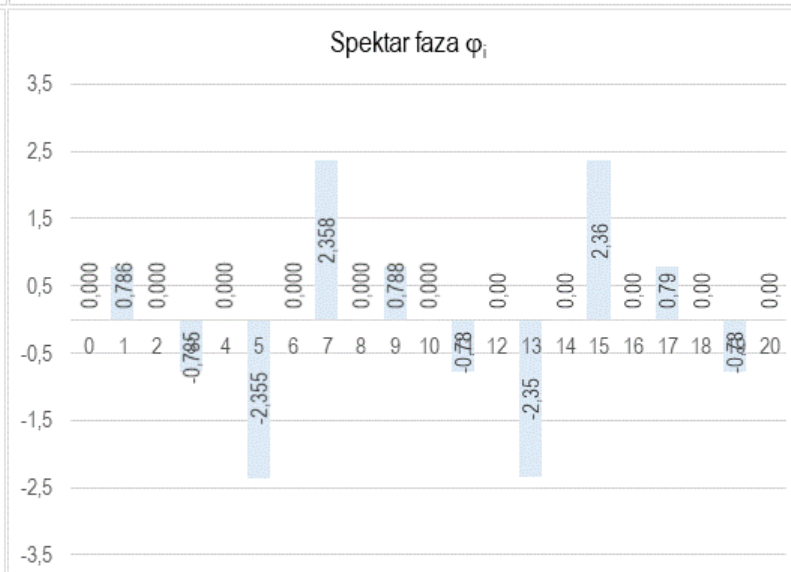
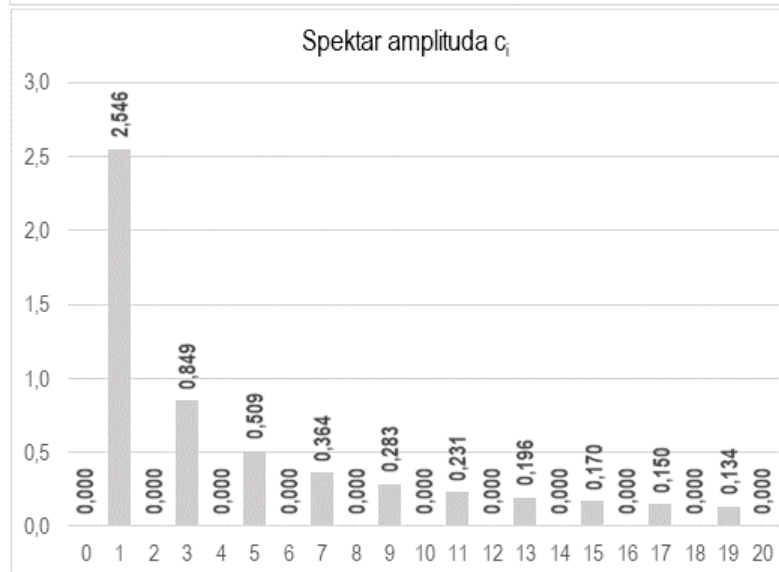
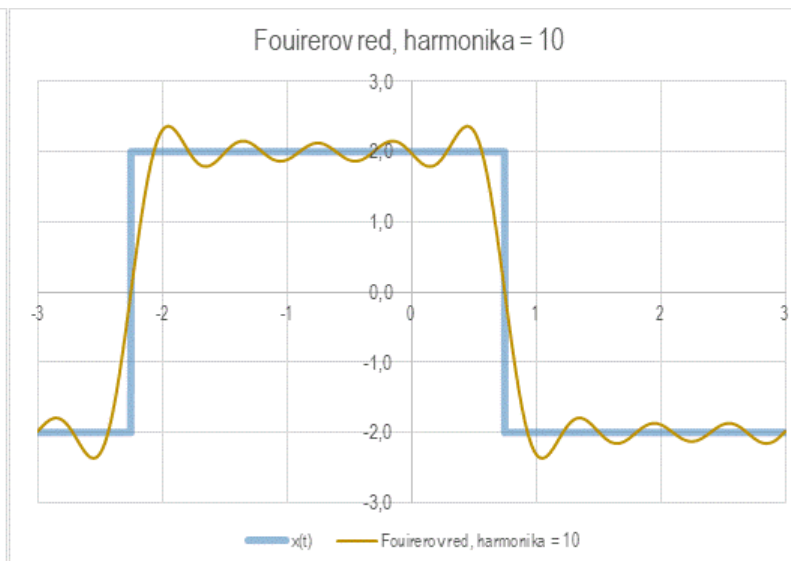
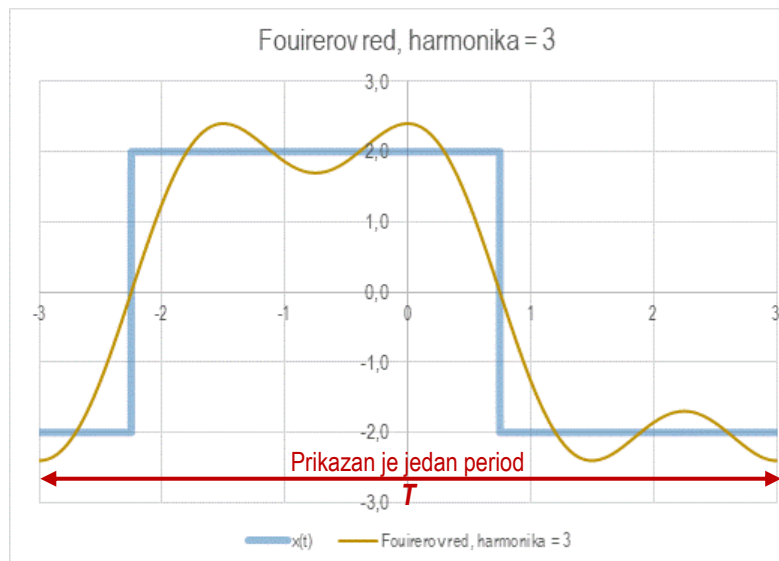
$$(2/T) \cdot \int x(t) \cdot \cos(4 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(4 \cdot \omega t) dt = 0$$



Primjer 3. Ni parna ni neparna funkcija

Primjer 3. Ni parna ni neparna funkcija



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 3.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
0,000	1,800	-1,801	0,000	0,000	0,601	0,600	0,000	0,000	-0,360	0,361	0,000	0,000	-0,258	-0,257	0,000	0,000	0,200	-0,201	0,000	0,000

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	0,000	2,546	0,000	0,849	0,000	0,509	0,000	0,364	0,000	0,283	0,000
$f_i =$	0,000	0,786	0,000	-0,785	0,000	-2,355	0,000	2,358	0,000	0,788	0,000

Zašto nema parnih harmonika?

Razlikuje li se spektar amplituda od Primjera 1. i 2.?

Razlikuje li se spektar faza Primjera 1. i 2.? („okno promatranja“)

Parsevalov teorem:

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 3,8384$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 4$$

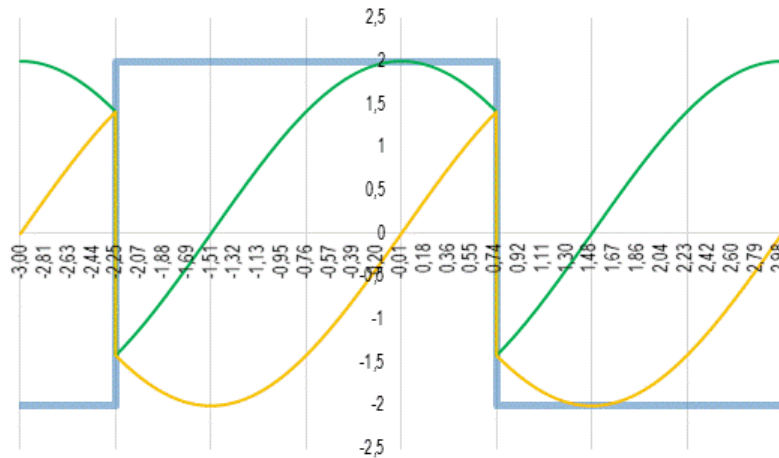
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 3.



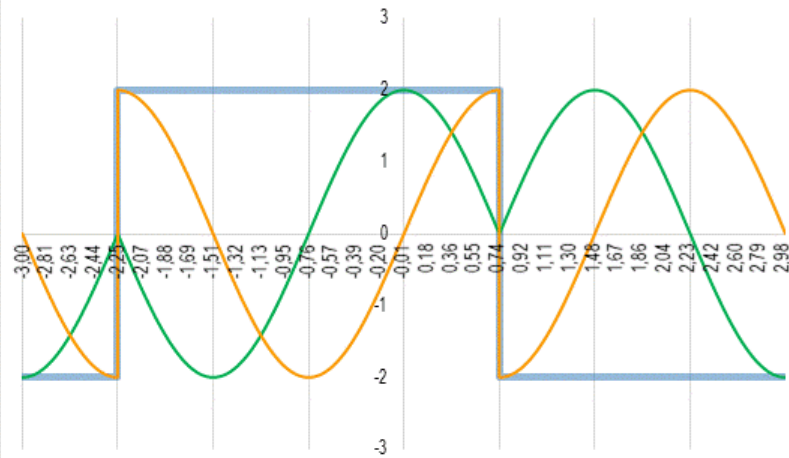
$$(2/T) \cdot \int x(t) \cdot \cos(1 \cdot \omega t) dt = 1,8$$

$$(2/T) \cdot \int x(t) \cdot \sin(1 \cdot \omega t) dt = -1,8$$



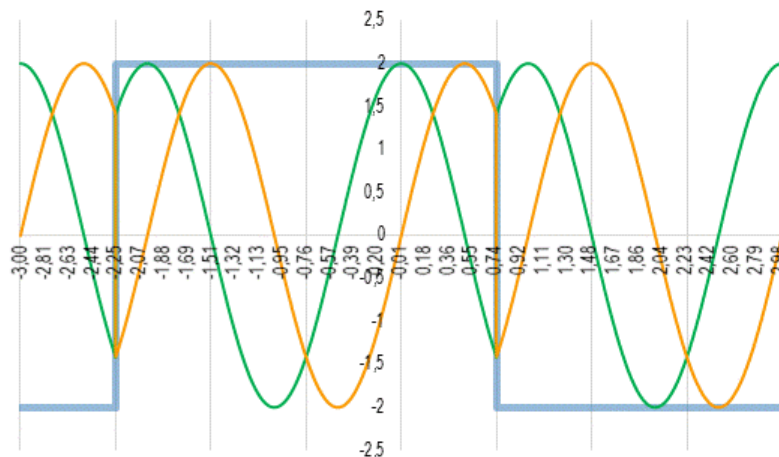
$$(2/T) \cdot \int x(t) \cdot \cos(2 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(2 \cdot \omega t) dt = 0$$



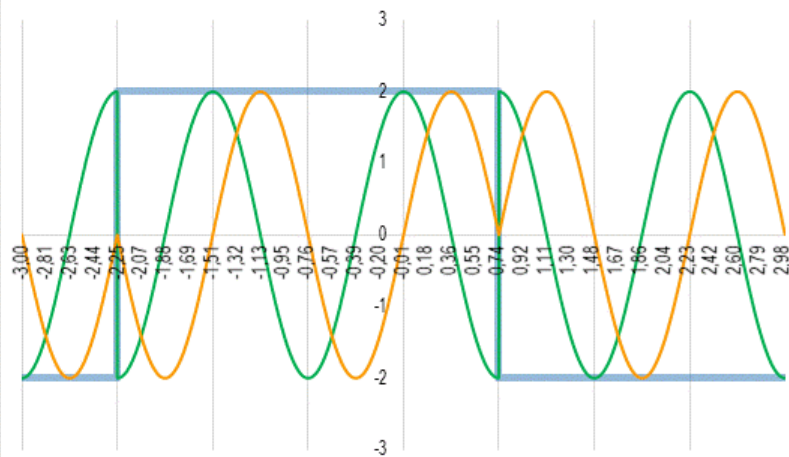
$$(2/T) \cdot \int x(t) \cdot \cos(3 \cdot \omega t) dt = 0,6$$

$$(2/T) \cdot \int x(t) \cdot \sin(3 \cdot \omega t) dt = 0,6$$



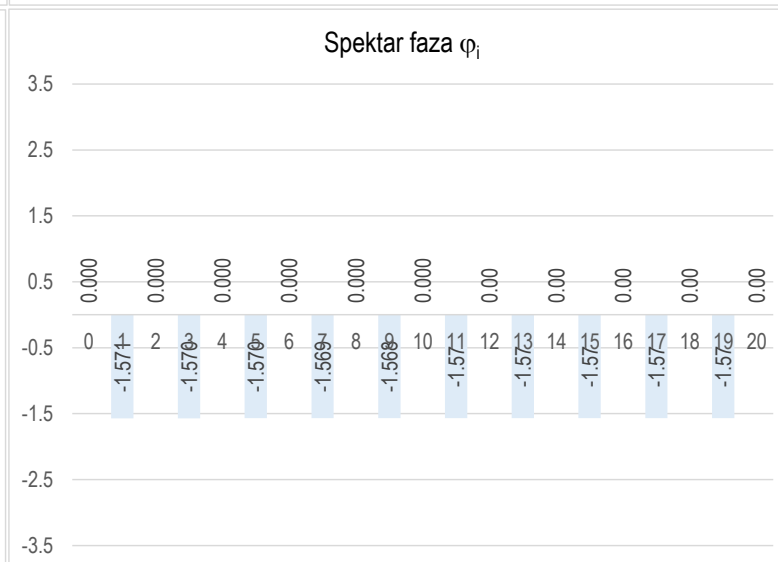
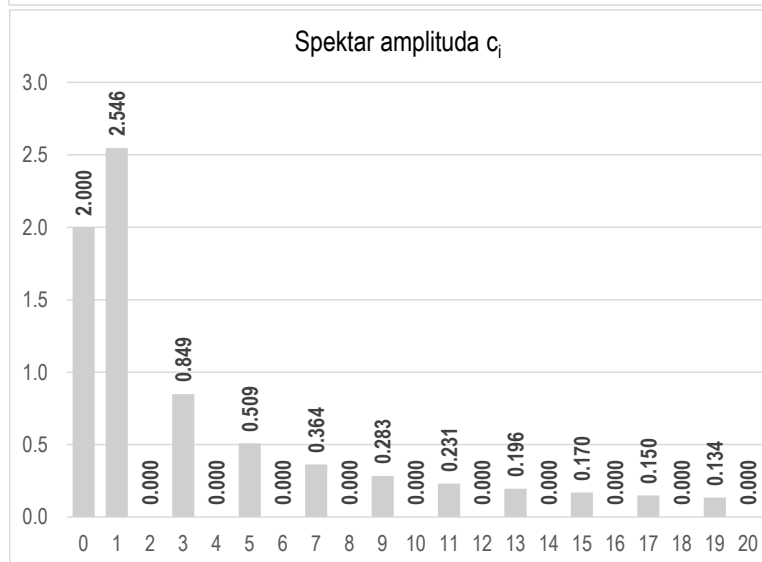
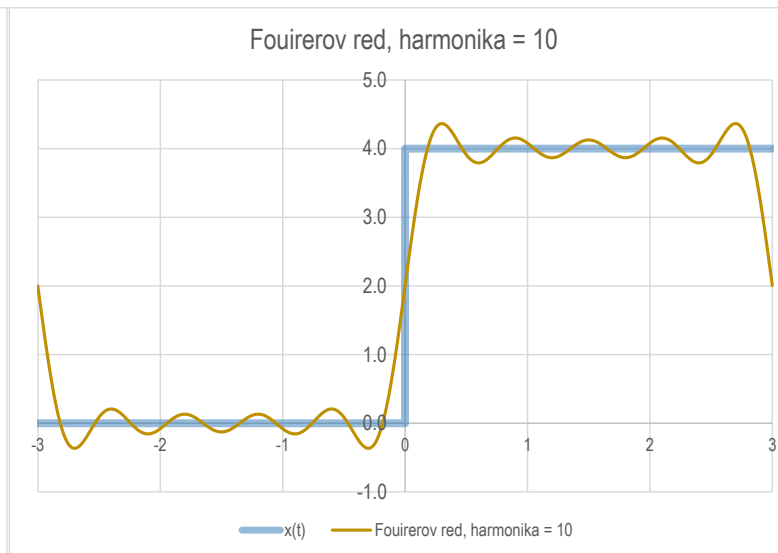
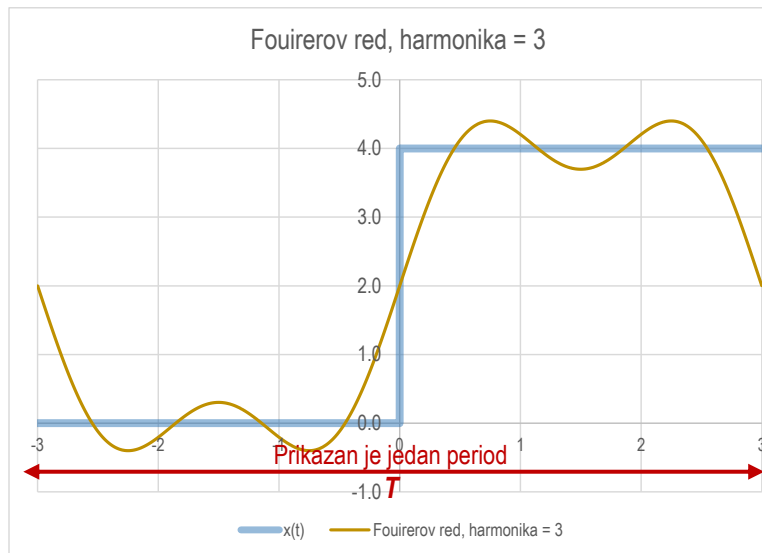
$$(2/T) \cdot \int x(t) \cdot \cos(4 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(4 \cdot \omega t) dt = 0$$



Primjer 4. Signal s istosmjernom komponentom

Primjer 4. Signal s istosmjernom komponentom



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 4.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
4,000	0,001	2,546	0,000	0,000	0,001	0,849	0,000	0,000	0,001	0,509	0,000	0,000	0,001	0,364	0,000	0,000	0,001	0,283	0,000	0,000

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	2,000	2,546	0,000	0,849	0,000	0,509	0,000	0,364	0,000	0,283	0,000
$f_{li} =$	0,000	-1,571	0,000	-1,570	0,000	-1,570	0,000	-1,569	0,000	-1,568	0,000

Parsevalov teorem:

$$a_0 = 4$$

$$U_{DC} = (a_0 / 2) = c_0$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$x(t) = c_0 + \sum_{i=1}^{+\infty} c_i \cos(i \cdot \omega_1 t + \varphi_i)$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 7,8384$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 8$$

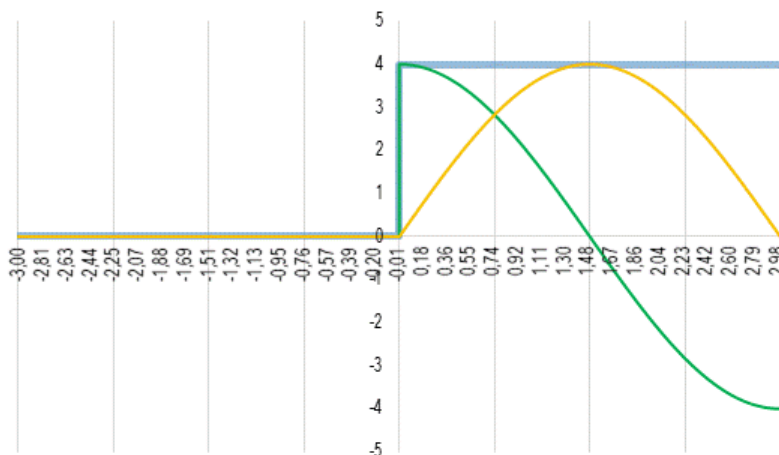
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 4.



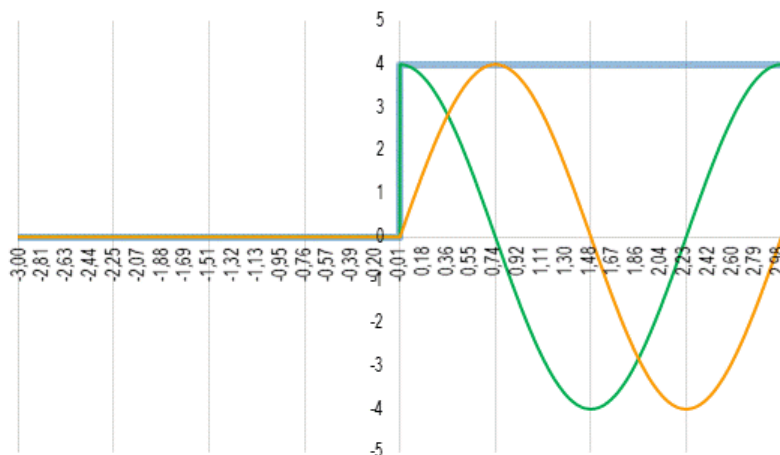
$$(2/T) \cdot \int x(t) \cdot \cos(1 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(1 \cdot \omega t) dt = 2,55$$



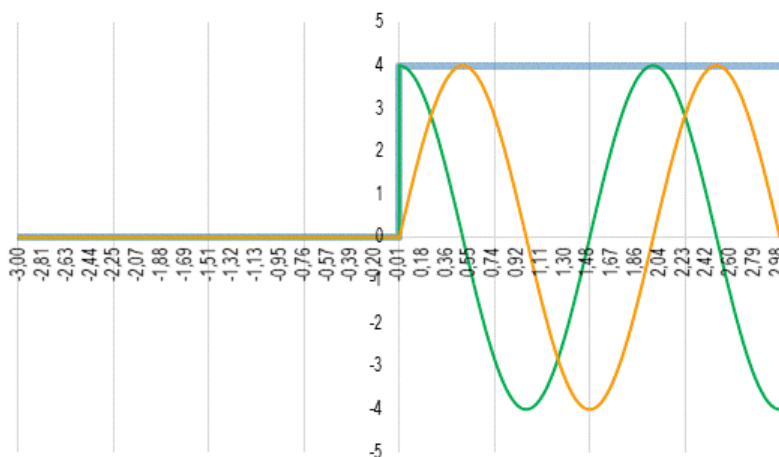
$$(2/T) \cdot \int x(t) \cdot \cos(2 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(2 \cdot \omega t) dt = 0$$



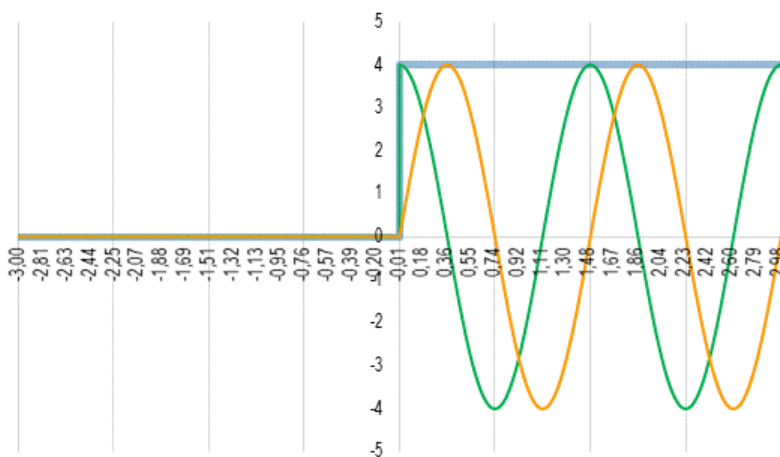
$$(2/T) \cdot \int x(t) \cdot \cos(3 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(3 \cdot \omega t) dt = 0,85$$

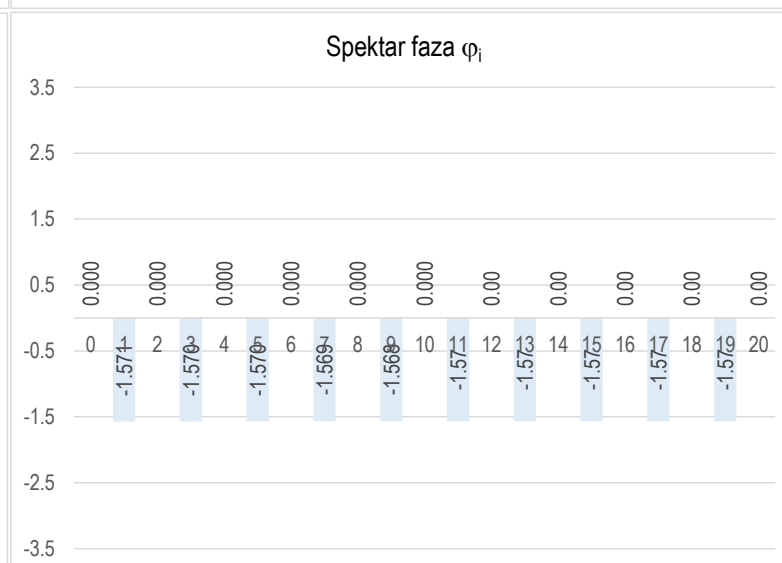
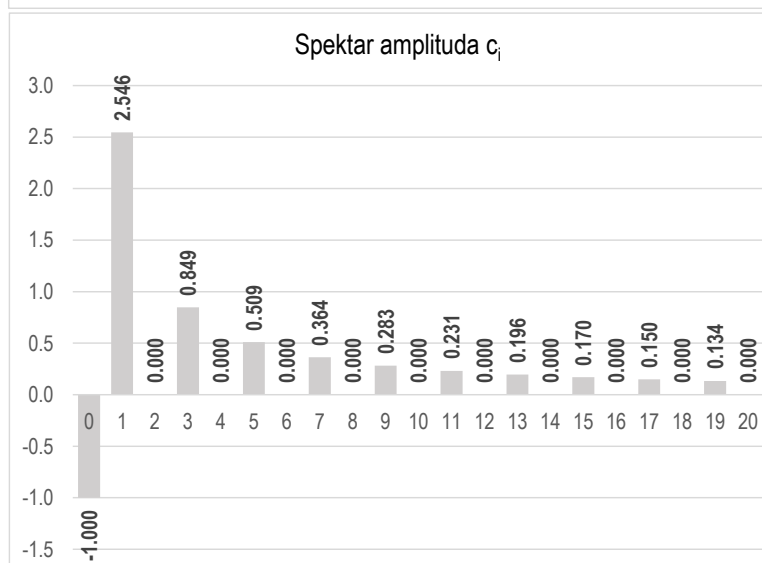
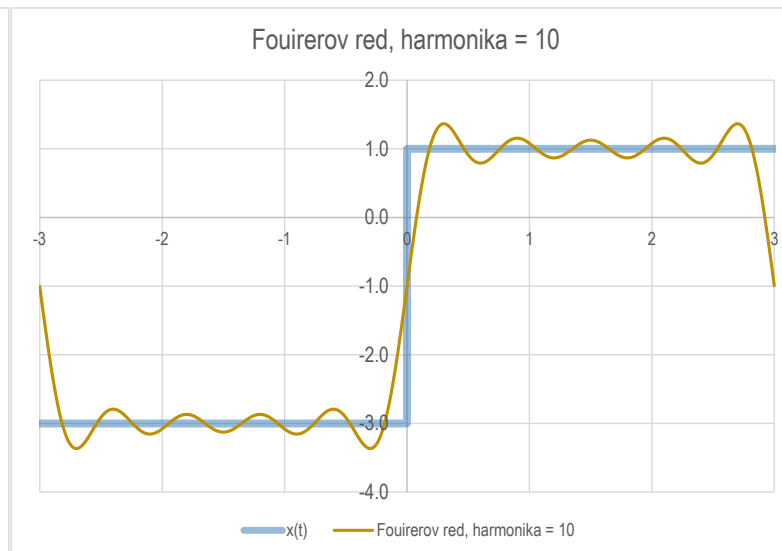
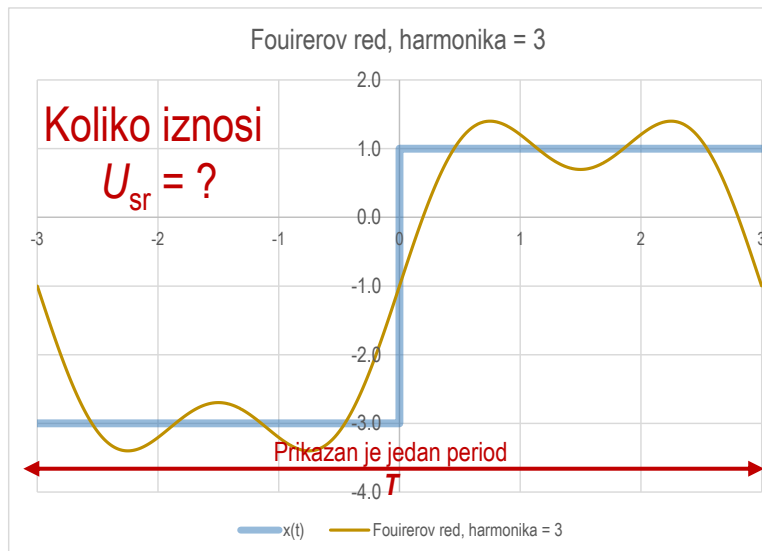


$$(2/T) \cdot \int x(t) \cdot \cos(4 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(4 \cdot \omega t) dt = 0$$



Primjer 5. Negativna istosmjerna komponenta



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 5.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
-2,000	0,000	2,546	0,000	0,000	0,000	0,849	0,000	0,000	0,000	0,509	0,000	0,000	0,000	0,364	0,000	0,000	0,000	0,283	0,000	0,000

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	-1,000	2,546	0,000	0,849	0,000	0,509	0,000	0,364	0,000	0,283	0,000
$f_{i_i} =$	0,000	-1,571	0,000	-1,570	0,000	-1,570	0,000	-1,569	0,000	-1,568	0,000

$$a_0 = -2$$

$$U_{DC} = (a_0 / 2) = c_0 = -1$$

$$x(t) = c_0 + \sum_{i=1}^{+\infty} c_i \cos(i \cdot \omega_1 t + \varphi_i)$$

Parsevalov teorem:

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

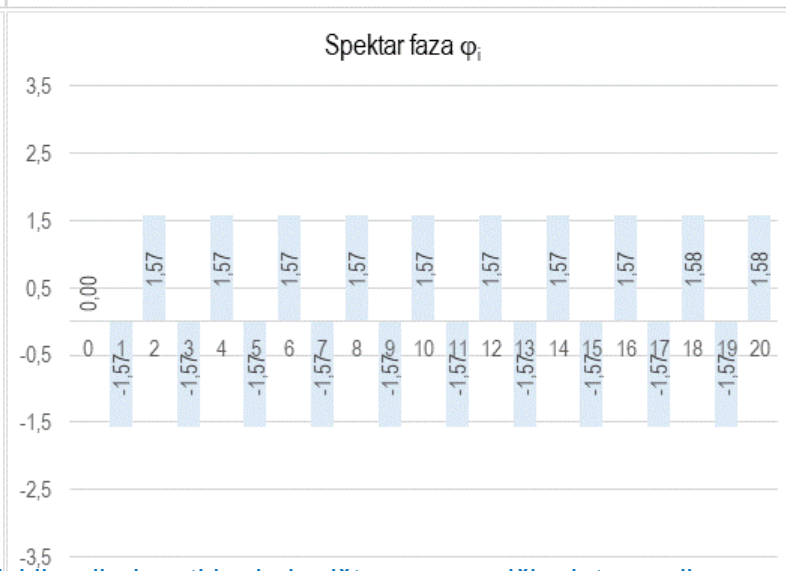
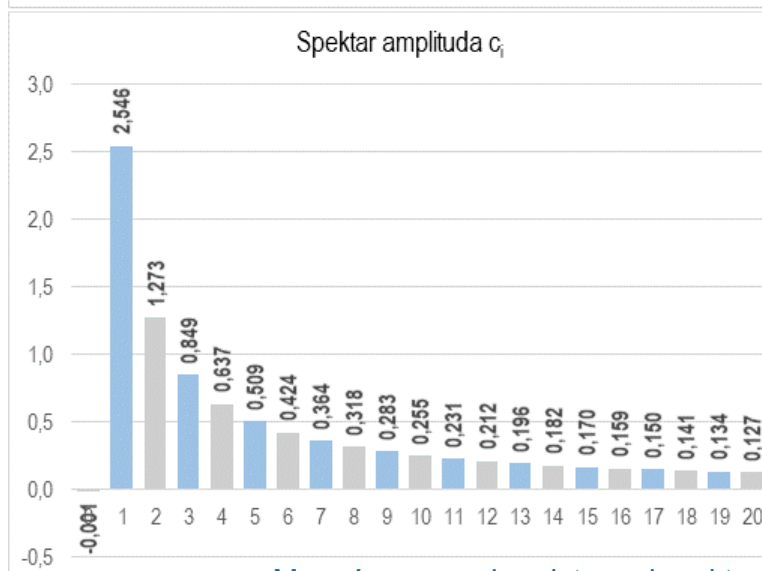
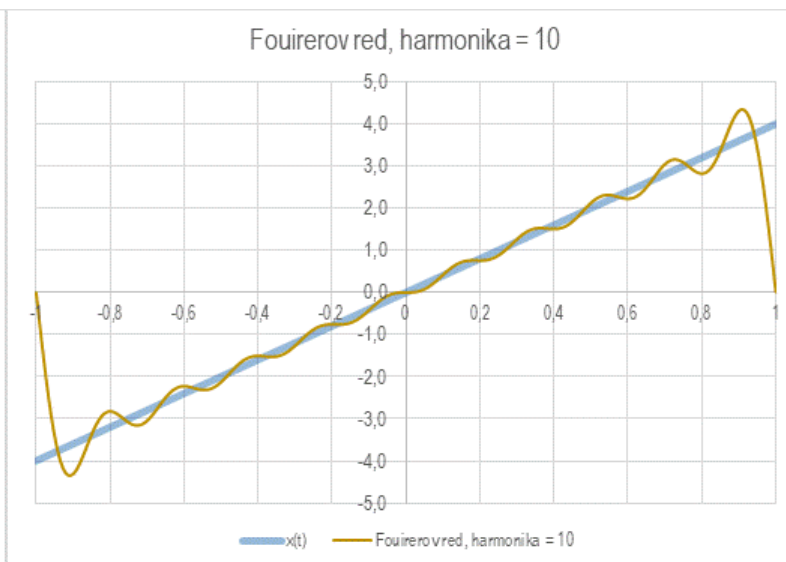
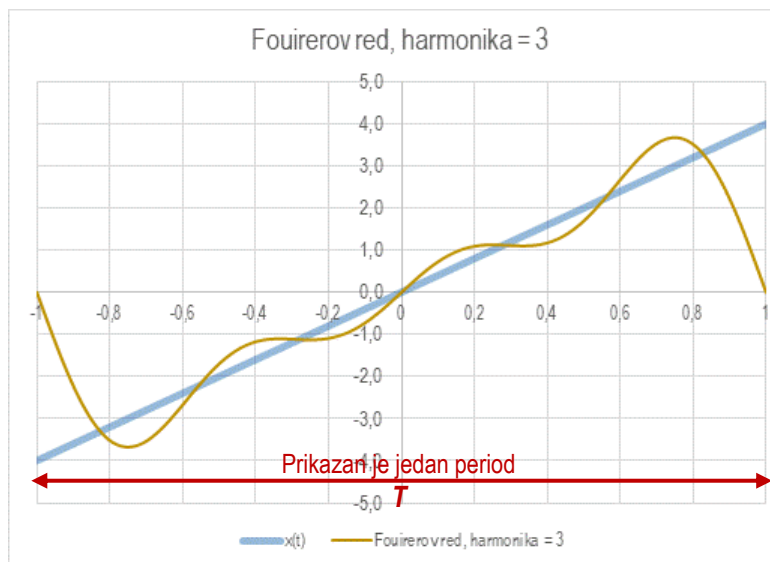
$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 4,8384$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 5$$

Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 6. Pilasti signal

Primjer 6. Neparni i parni harmonici



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 6.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
0,000	0,000	2,546	0,000	-1,273	0,000	0,849	0,000	-0,637	0,000	0,509	0,000	-0,424	0,000	0,364	0,000	-0,318	0,000	0,283	0,000	-0,255

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	0,0000	2,5465	1,2732	0,8488	0,6366	0,5093	0,4244	0,3638	0,3183	0,2829	0,2546
$f_{i_j} =$	0,00	-1,57	1,57	-1,57	1,57	-1,57	1,57	-1,57	1,57	-1,57	1,57

Parsevalov teorem:

Usporedite amplitude neparnih harmonika sa spektrom pravokutnog signala iz Primjera 1.

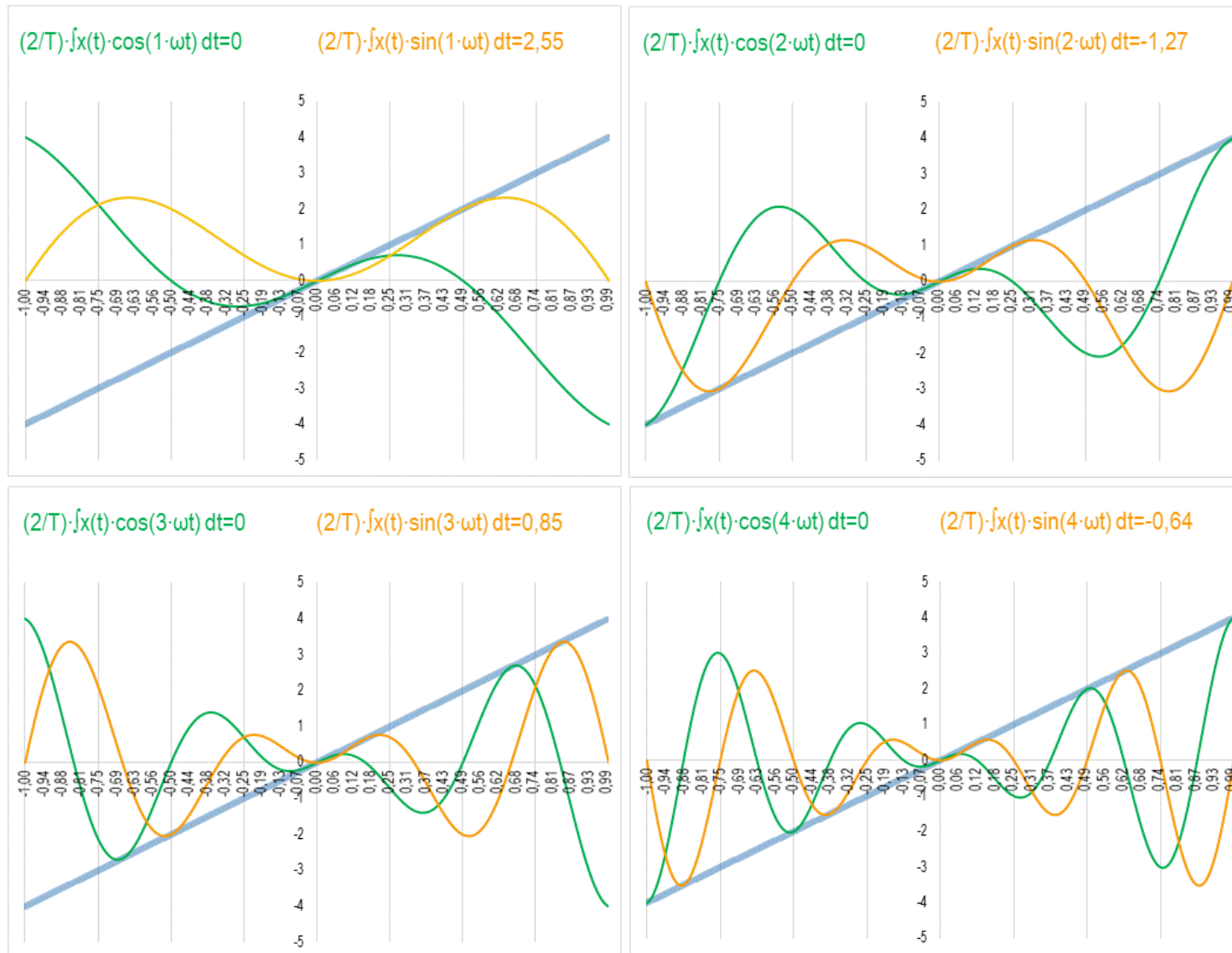
$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 5,02478$$

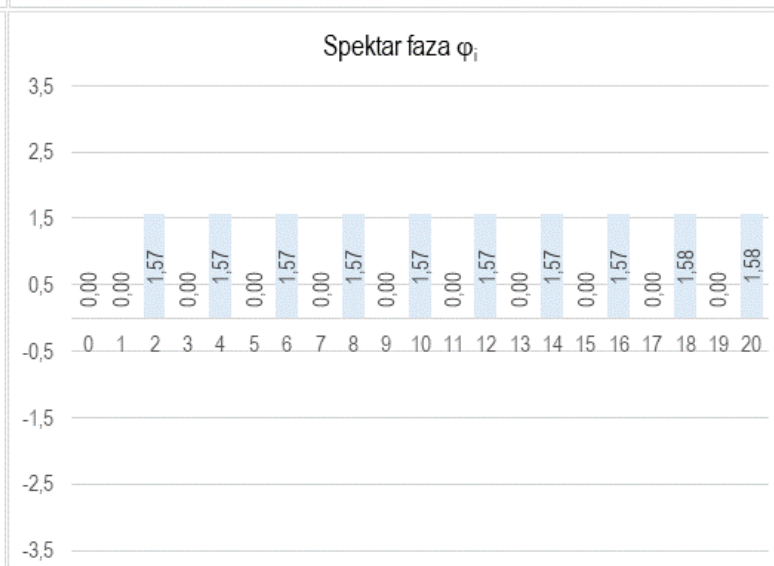
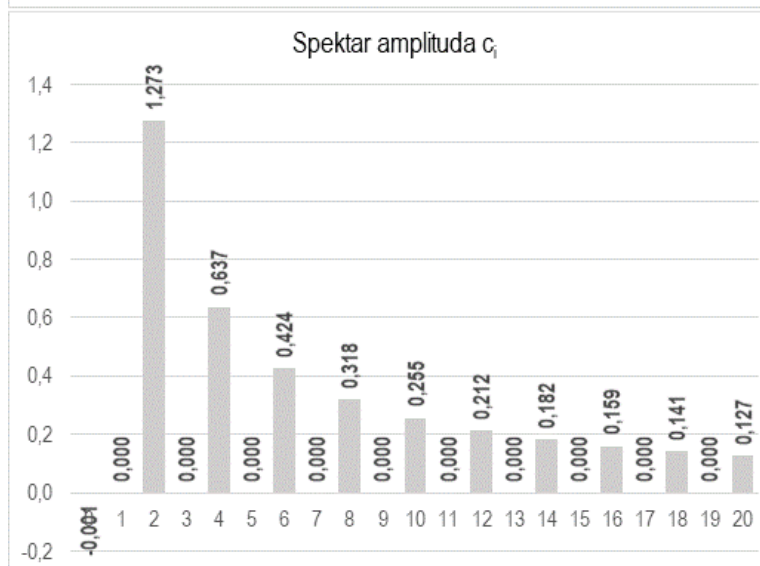
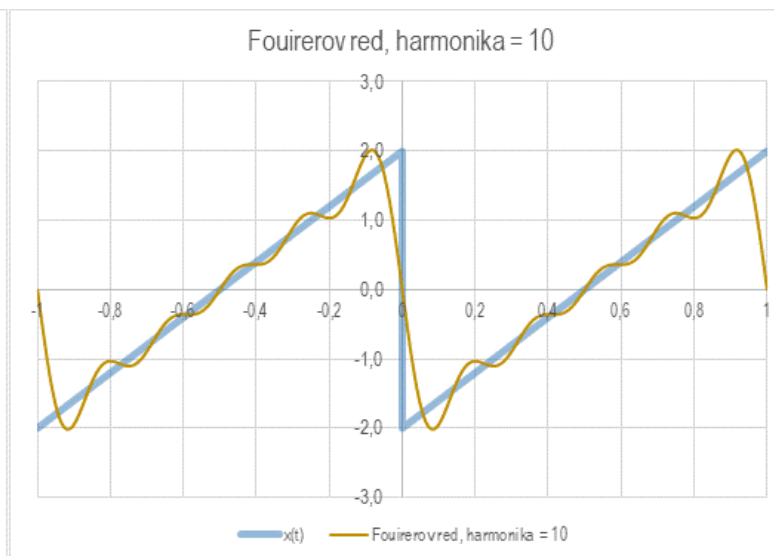
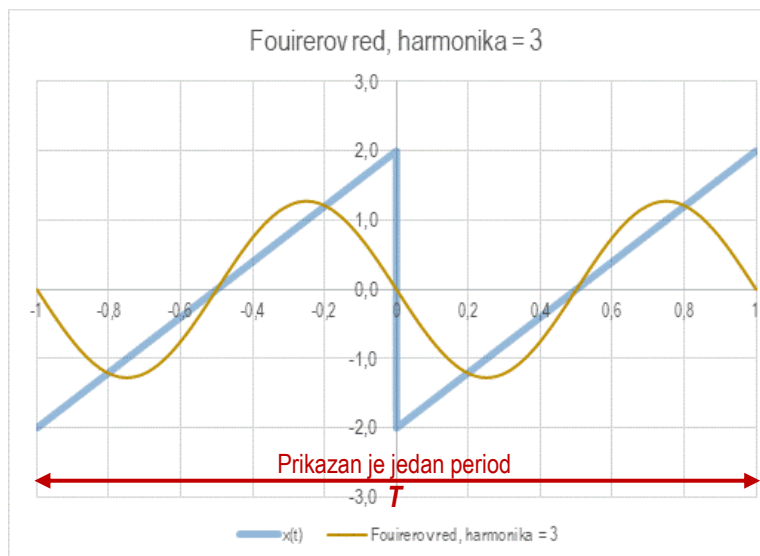
$$\frac{1}{T} \int_T [x(t)]^2 dt = 5,33333$$

Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 6.



Primjer 6a. [Pilasti (Primjer 6.) – Pravokutni signal (Primjer 1.)]



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 6a.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
0,000	0,000	0,000	0,000	-1,273	0,000	0,000	0,000	-0,637	0,000	0,000	0,000	-0,424	0,000	0,000	0,000	-0,318	0,000	0,000	0,000	-0,255

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	0,000	0,000	1,273	0,000	0,637	0,000	0,424	0,000	0,318	0,000	0,255
$f_{i_i} =$	0,000	0,000	1,571	0,000	1,572	0,000	1,572	0,000	1,573	0,000	1,573

Parsevalov teorem:

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 1,18636$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 1,33333$$

Signal koji je **razlika** pilastog signala iz
Primjera 6. i pravokutnog signala iz
Primjera 1. $\Rightarrow$ ima samo **parne harmonike!**

Koja je **frekvencija** signala?

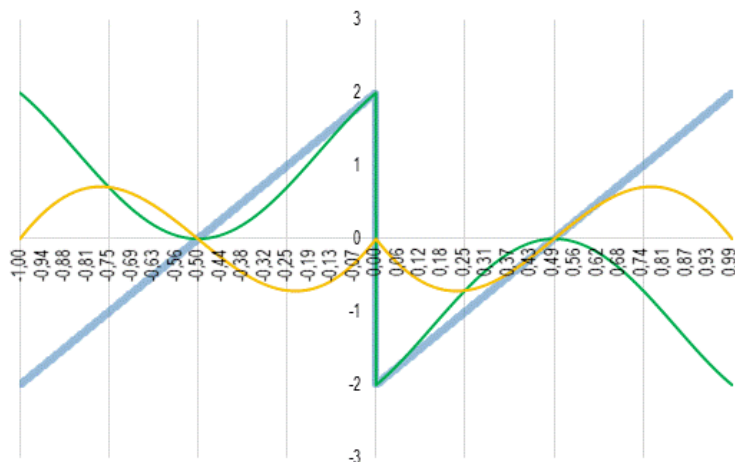
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 6a.



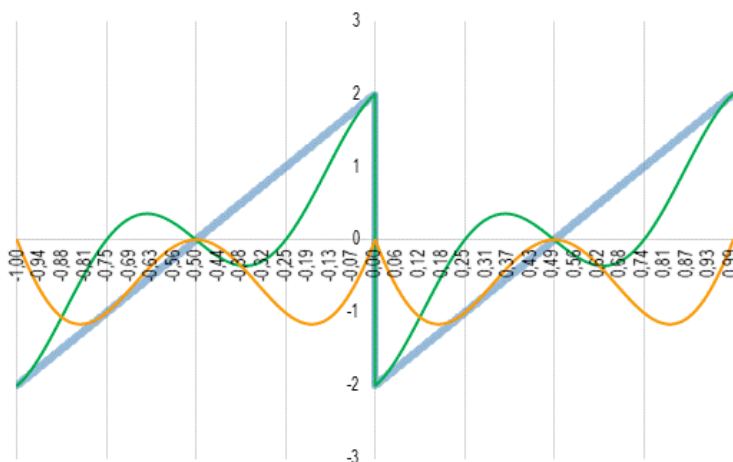
$$(2/T) \cdot \int x(t) \cdot \cos(1 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(1 \cdot \omega t) dt = 0$$



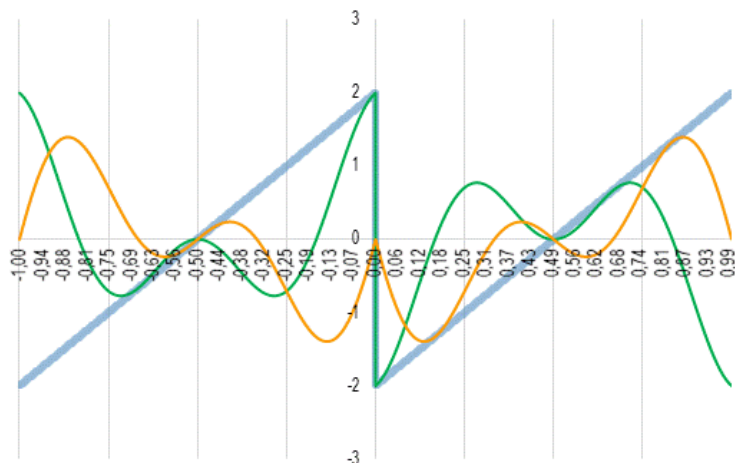
$$(2/T) \cdot \int x(t) \cdot \cos(2 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(2 \cdot \omega t) dt = -1,27$$



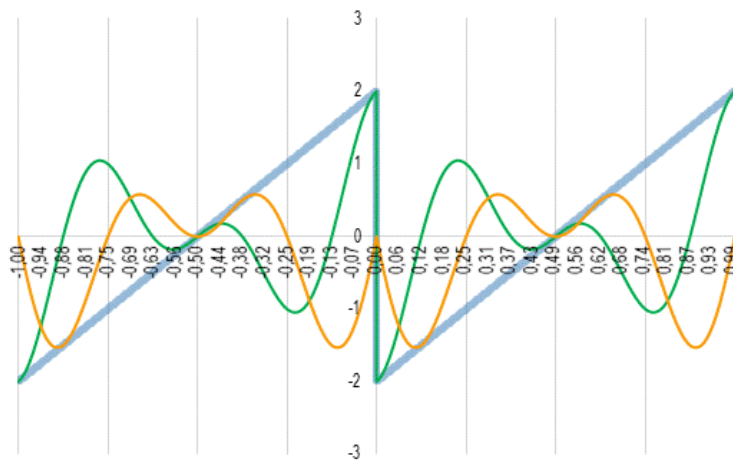
$$(2/T) \cdot \int x(t) \cdot \cos(3 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(3 \cdot \omega t) dt = 0$$



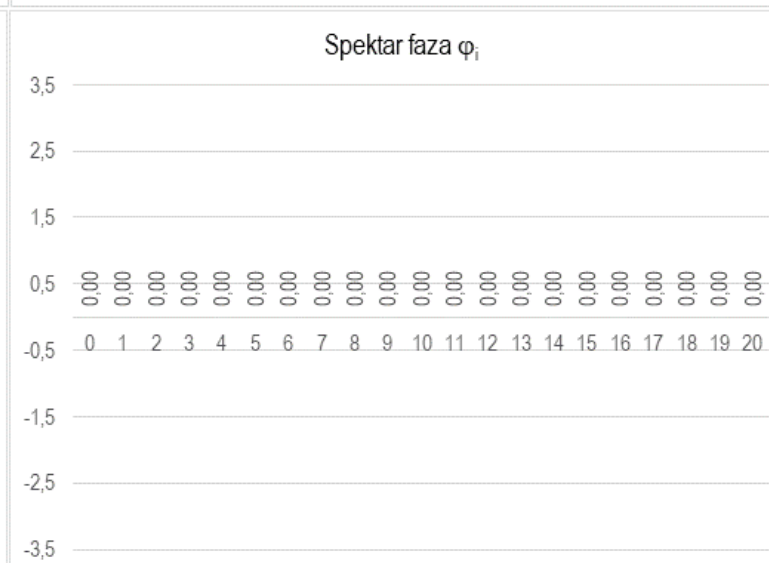
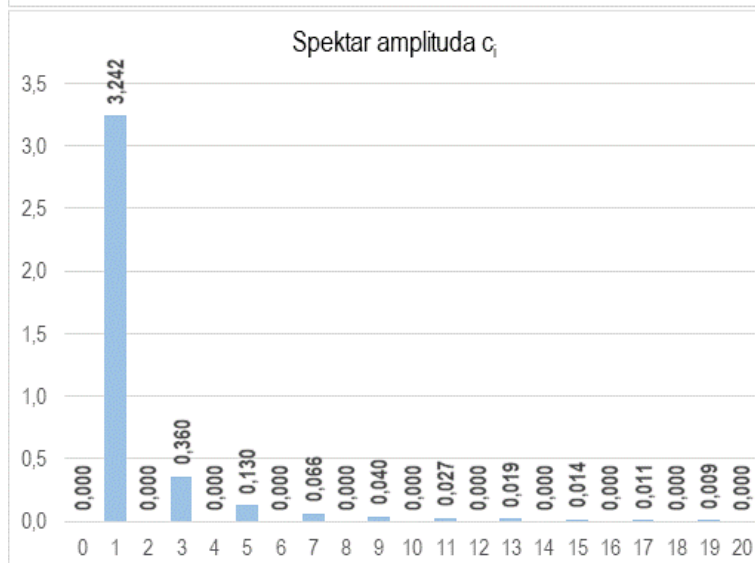
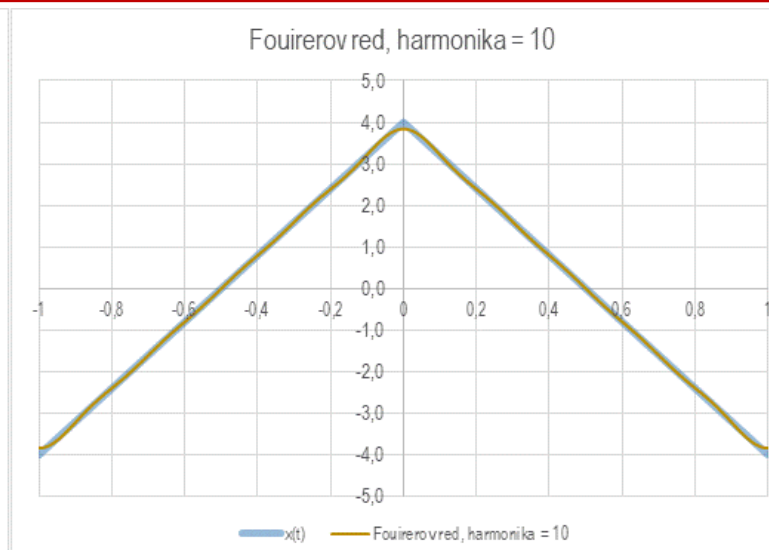
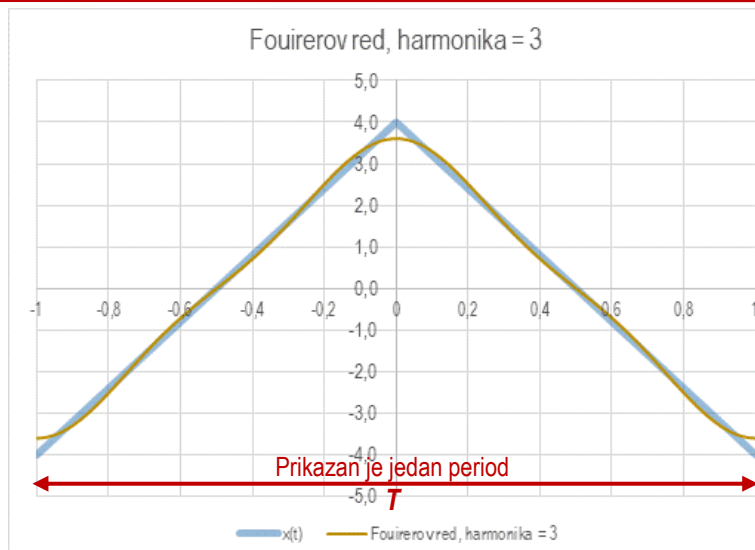
$$(2/T) \cdot \int x(t) \cdot \cos(4 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(4 \cdot \omega t) dt = -0,64$$



Primjer 7. Trokutasti signal

Primjer 7.



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 7.



$a_0=$	$a_1=$	$b_1=$	$a_2=$	$b_2=$	$a_3=$	$b_3=$	$a_4=$	$b_4=$	$a_5=$	$b_5=$	$a_6=$	$b_6=$	$a_7=$	$b_7=$	$a_8=$	$b_8=$	$a_9=$	$b_9=$	$a_{10}=$	$b_{10}=$
0,000	3,242	0,000	0,000	0,000	0,360	0,000	0,000	0,000	0,130	0,000	0,000	0,000	0,066	0,000	0,000	0,000	0,040	0,000	0,000	0,000

$i =$	0	1	2	3	4	5	6	7	8	9	10
$c_i =$	0,0000	3,2423	0,0000	0,3603	0,0000	0,1297	0,0000	0,0662	0,0000	0,0400	0,0000
$f_{i_j} =$	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

Parsevalov teorem:

Zašto su $b_i = 0$

Zašto nema parnih harmonika?

Zašto je faza $\varphi_1 = 0$?

Zašto c_i brzo padaju za veće i

Parsevalov teorem: već s 3. harmonikom greška je samo oko -0,2% $\Rightarrow$ usporedite s Primjerom 6.

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{+\infty} (a_i^2 + b_i^2) = \frac{1}{T} \int_T [x(t)]^2 dt$$

$$\left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{i=1}^{10} (a_i^2 + b_i^2) = 5,33247$$

$$\frac{1}{T} \int_T [x(t)]^2 dt = 5,33333$$

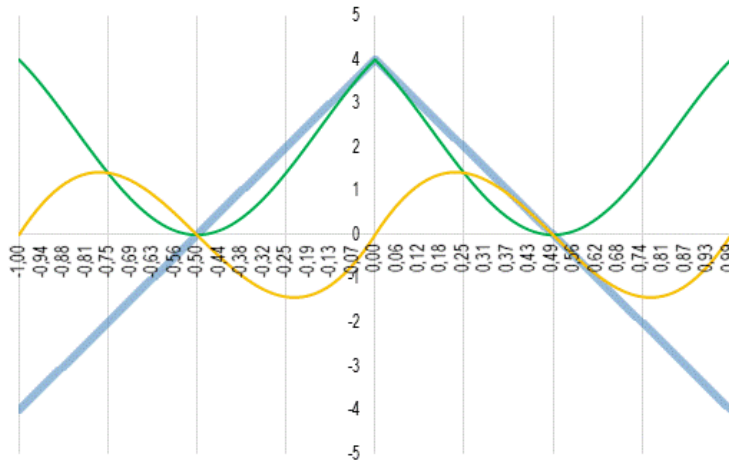
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 7.



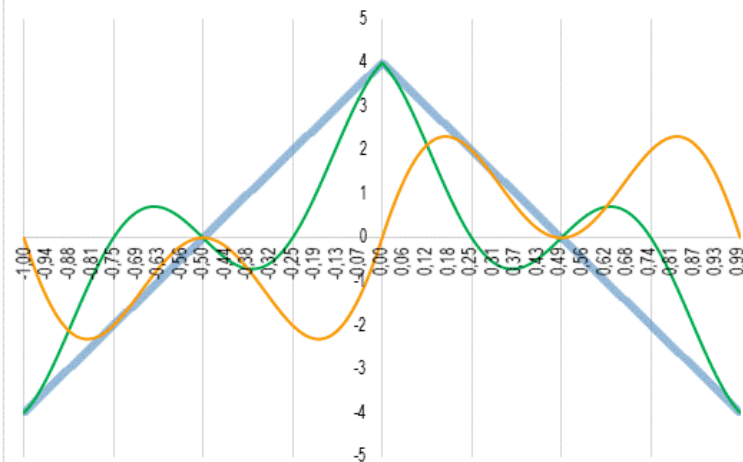
$$(2/T) \cdot \int x(t) \cdot \cos(1 \cdot \omega t) dt = 3,24$$

$$(2/T) \cdot \int x(t) \cdot \sin(1 \cdot \omega t) dt = 0$$



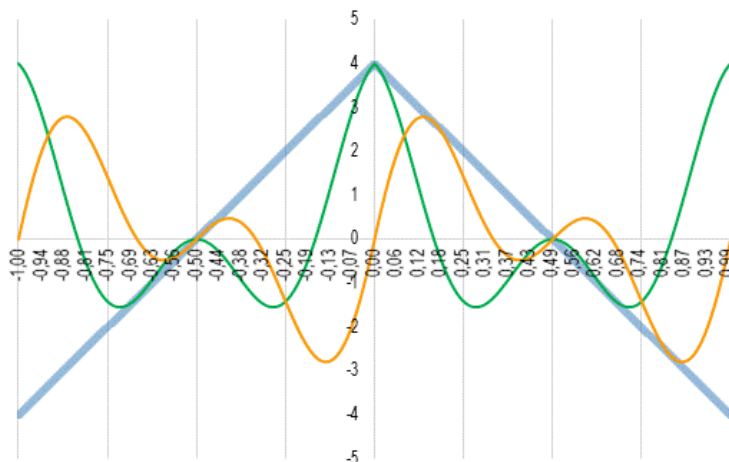
$$(2/T) \cdot \int x(t) \cdot \cos(2 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(2 \cdot \omega t) dt = 0$$



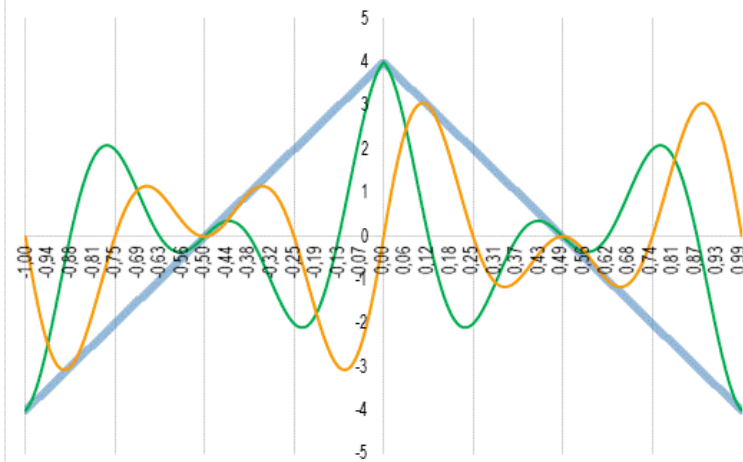
$$(2/T) \cdot \int x(t) \cdot \cos(3 \cdot \omega t) dt = 0,36$$

$$(2/T) \cdot \int x(t) \cdot \sin(3 \cdot \omega t) dt = 0$$



$$(2/T) \cdot \int x(t) \cdot \cos(4 \cdot \omega t) dt = 0$$

$$(2/T) \cdot \int x(t) \cdot \sin(4 \cdot \omega t) dt = 0$$

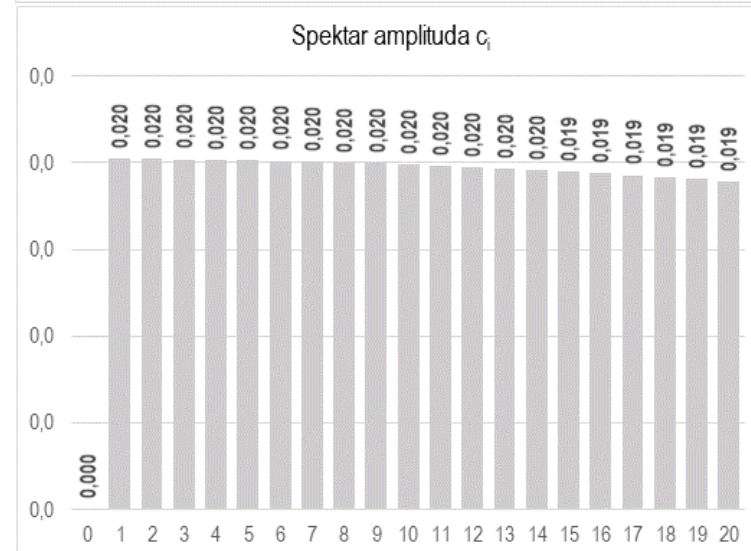
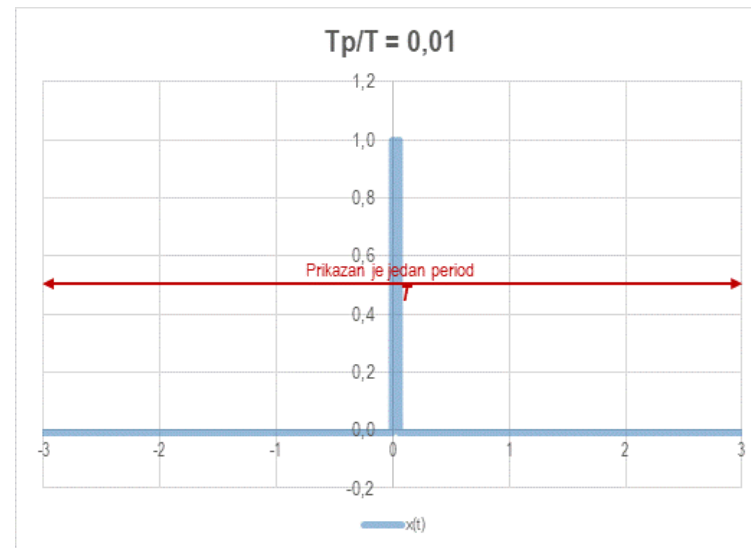
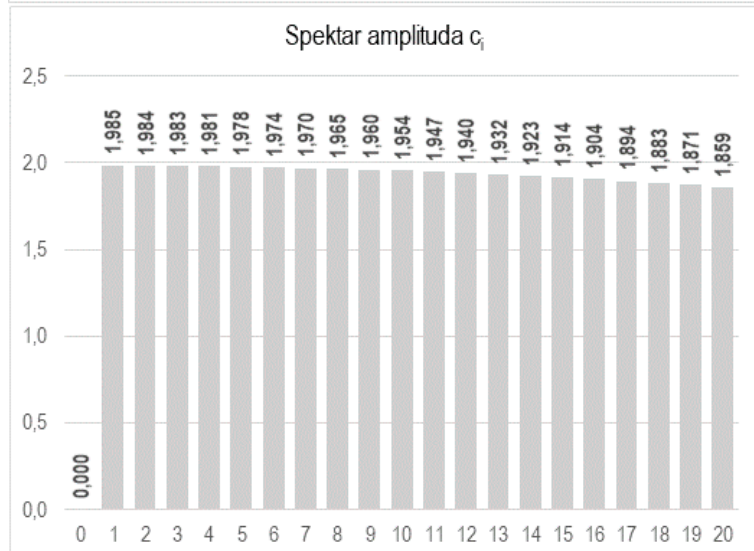
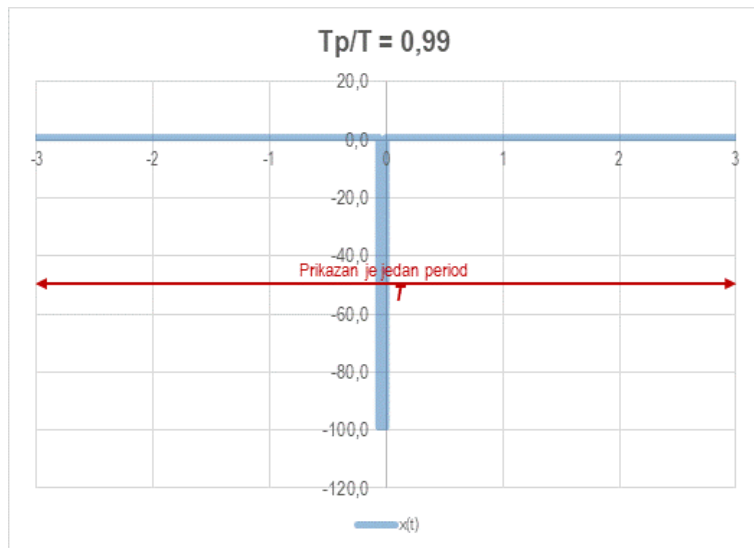


Primjer 8.

Utjecaj širine pravokutnog signala na amplitudni spektar signala

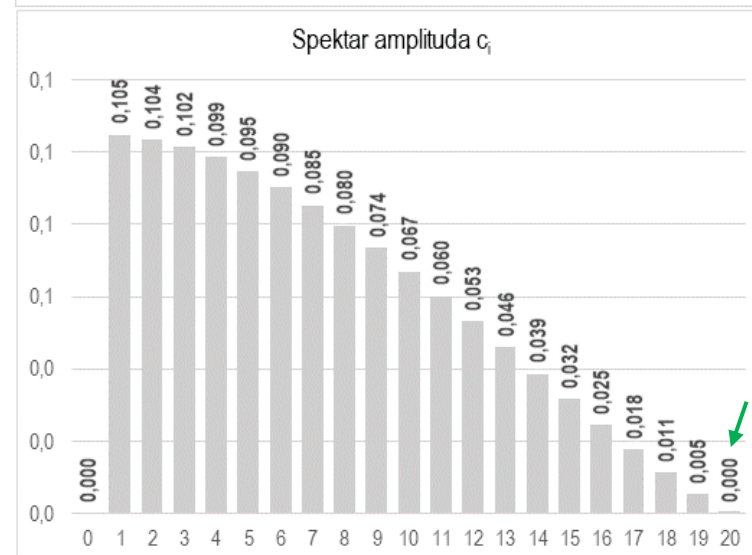
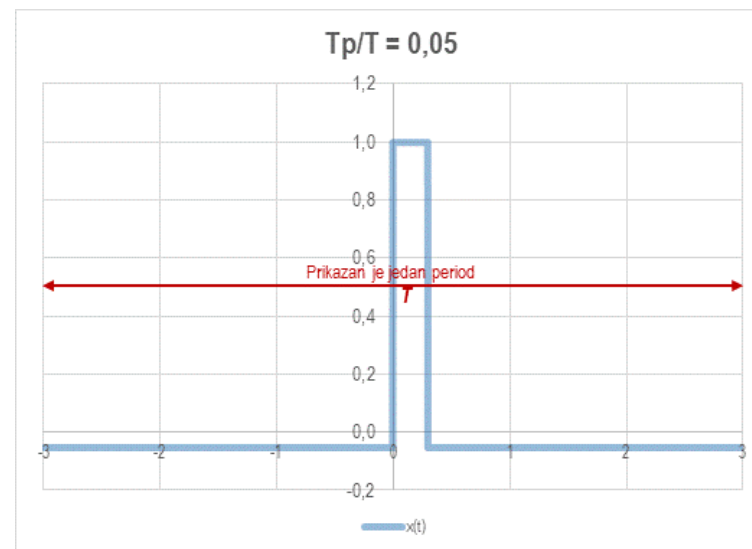
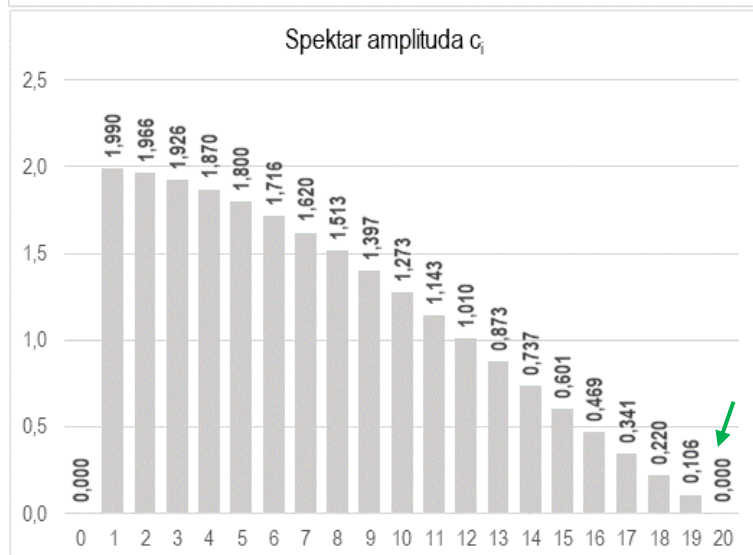
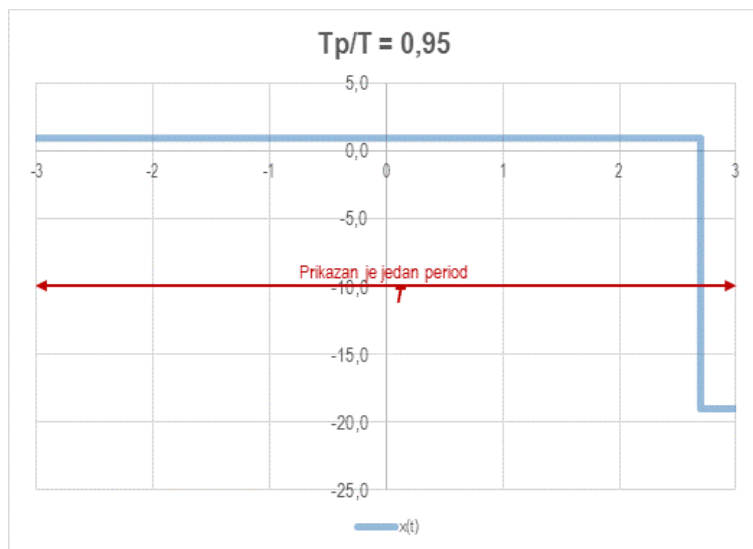
$$U_{DC} = 0$$

Primjer 8.



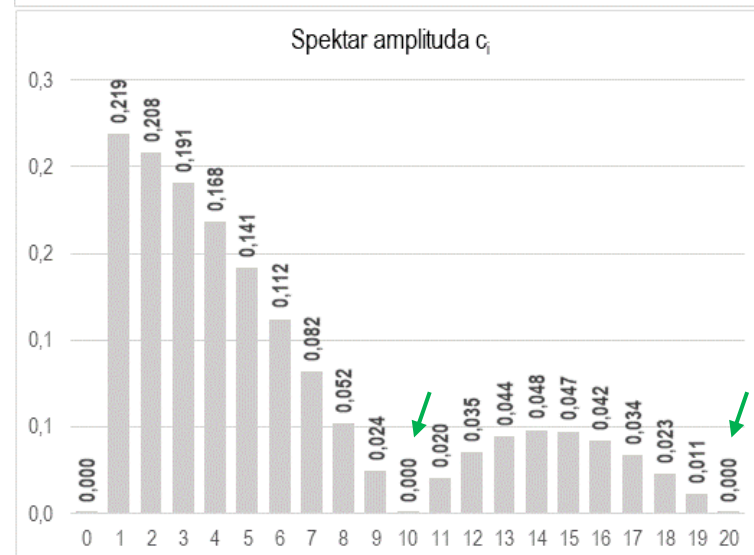
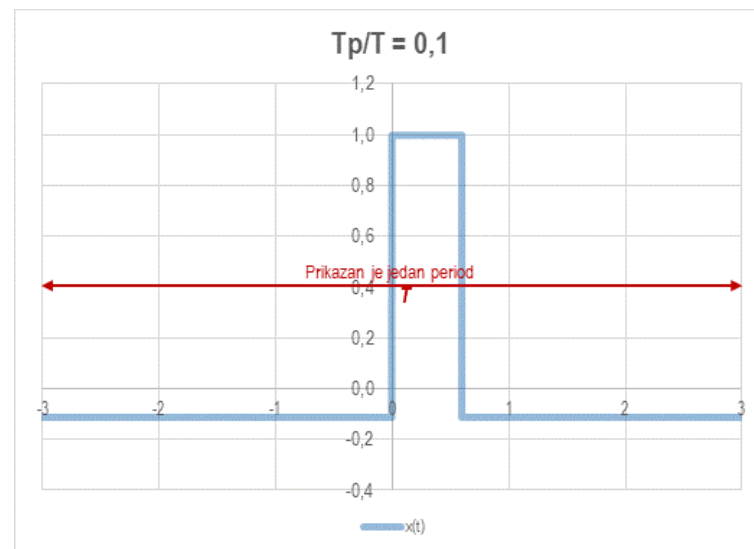
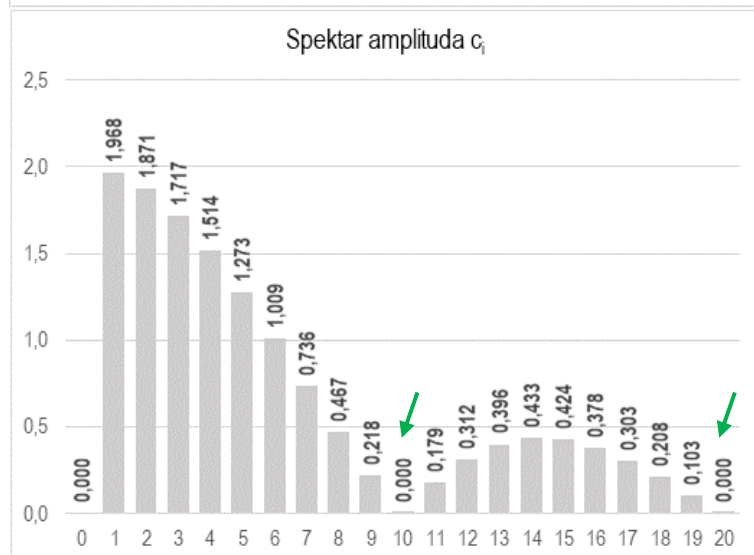
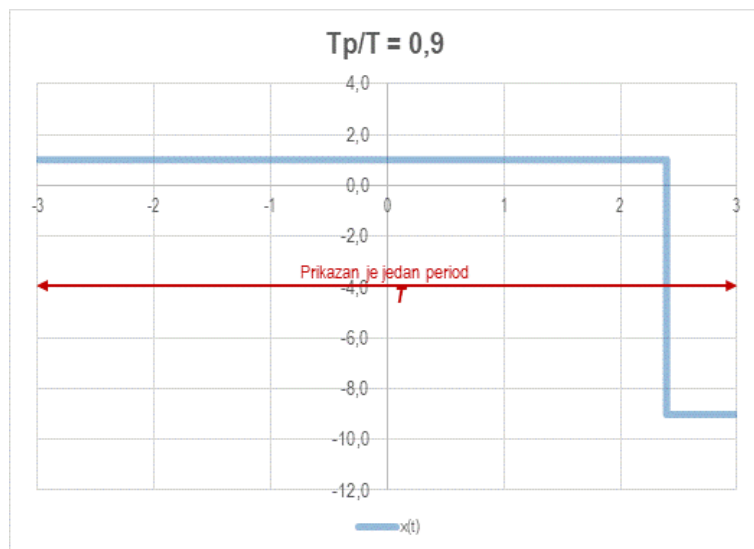
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8.



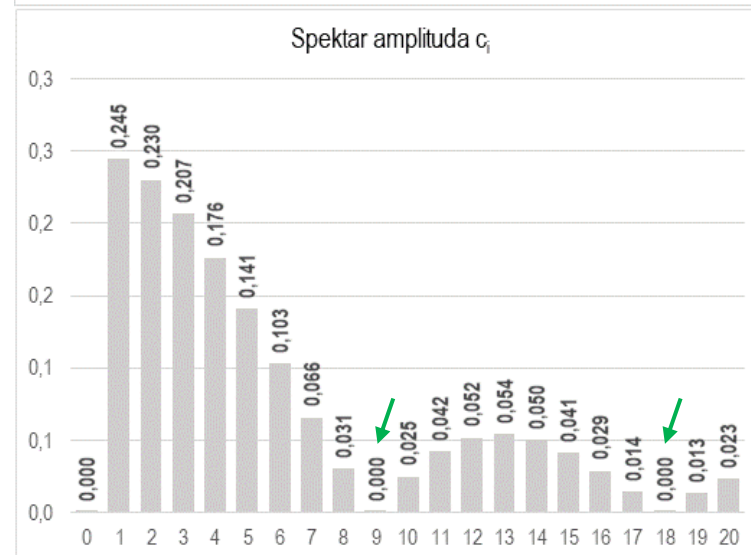
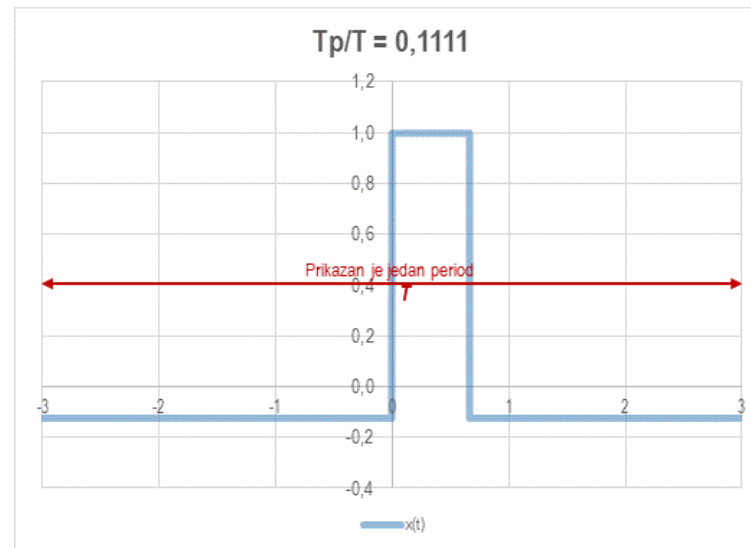
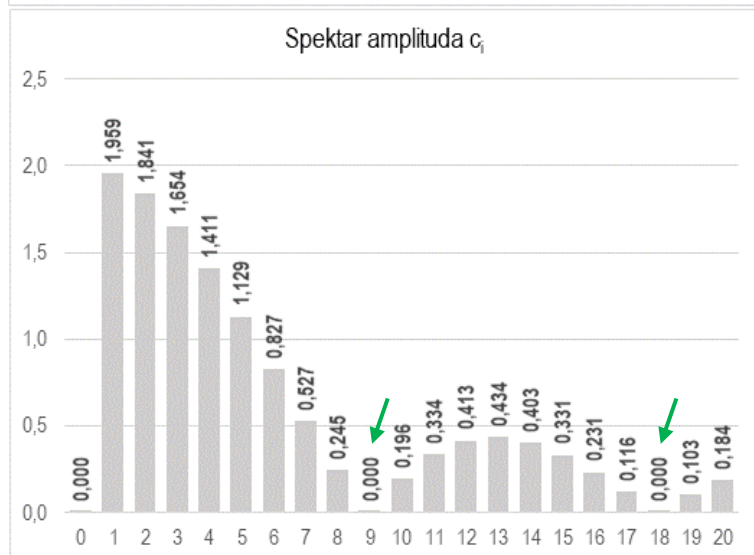
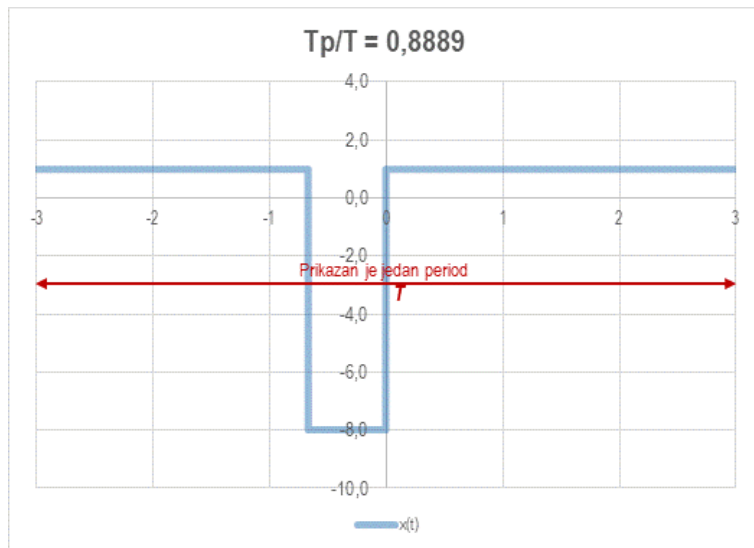
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8.



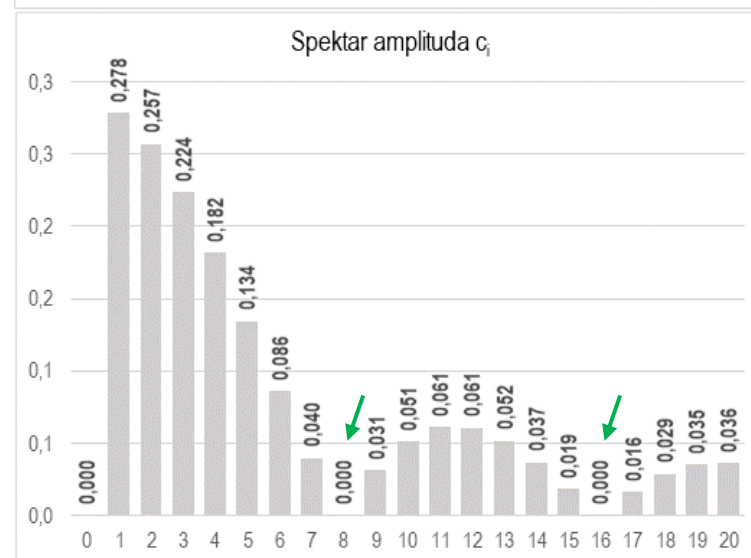
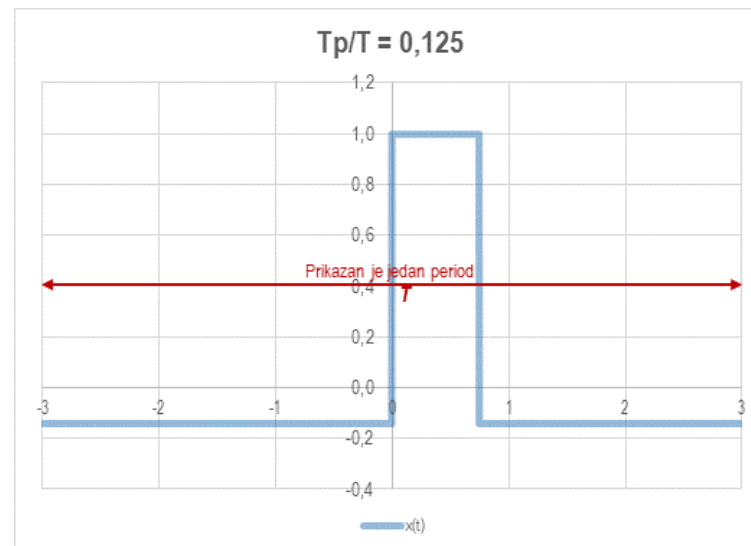
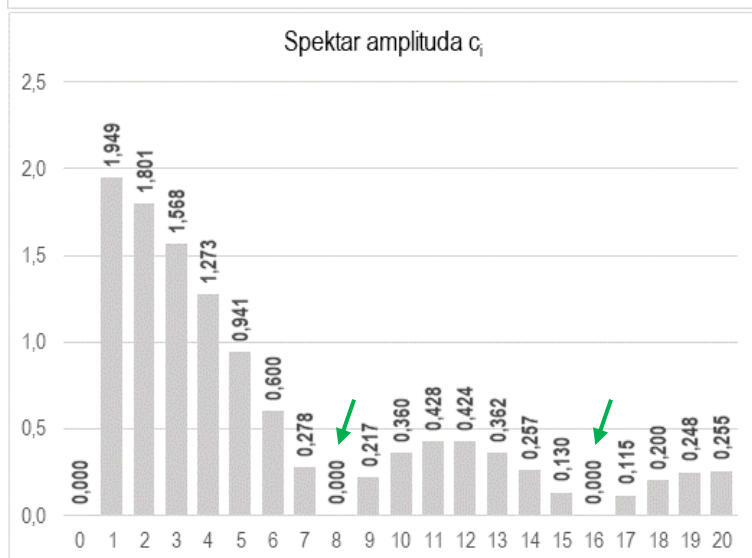
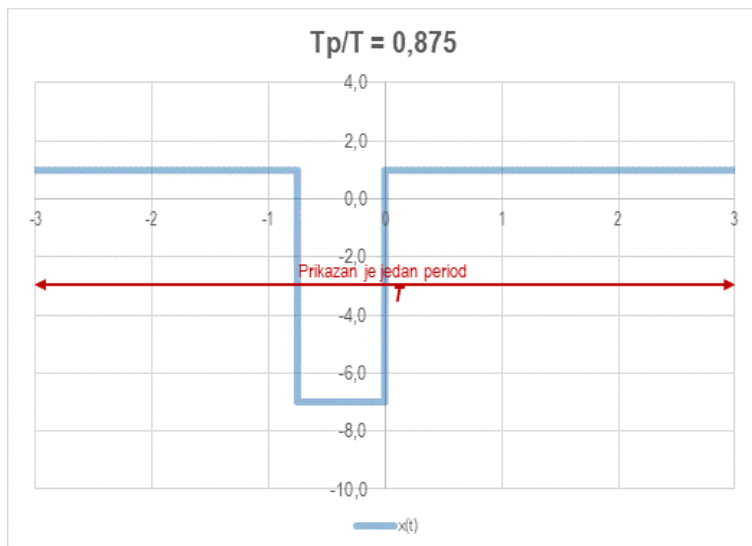
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Primjer 8.



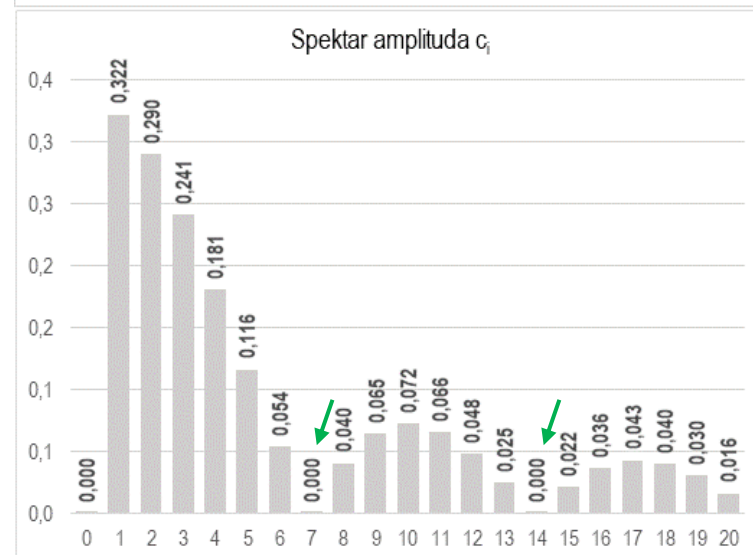
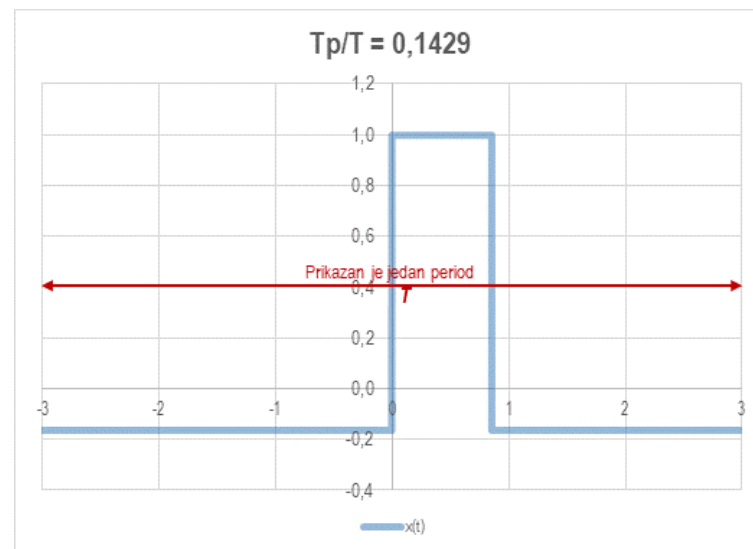
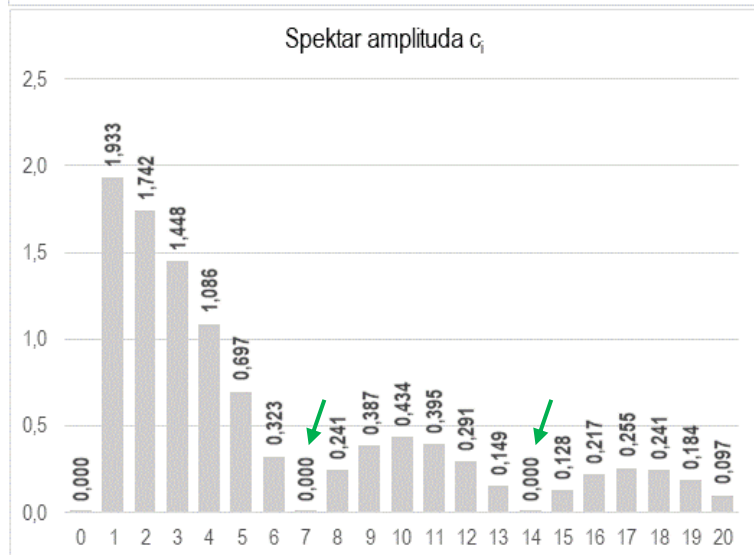
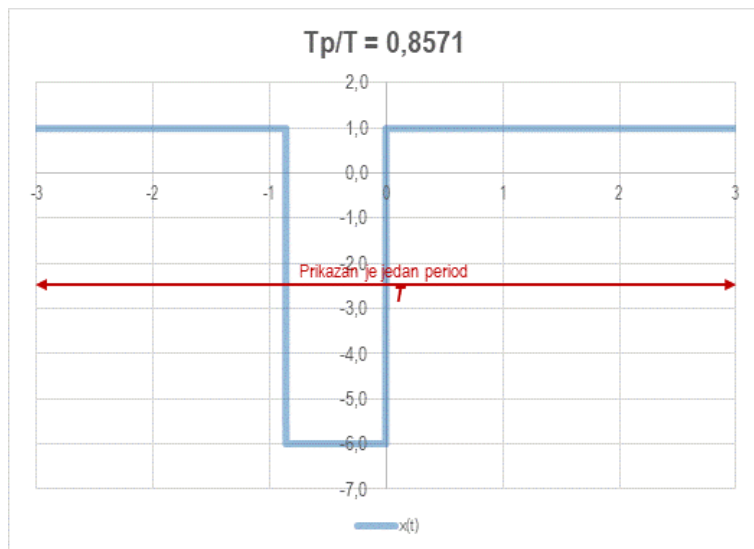
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8.



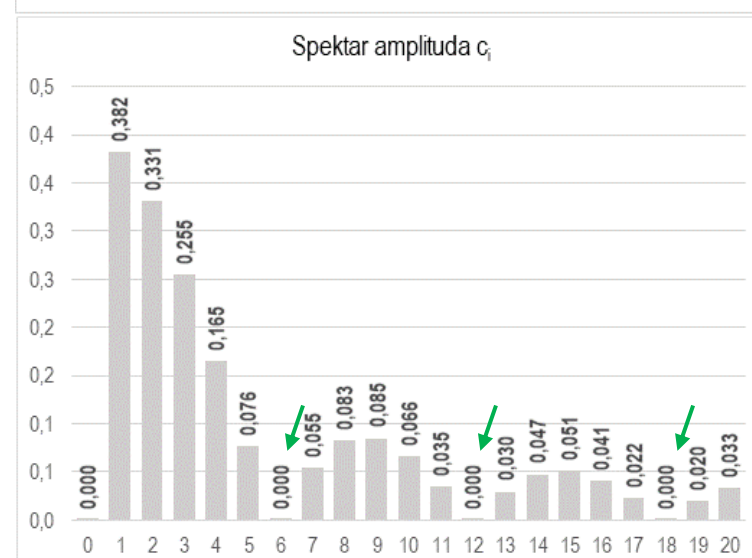
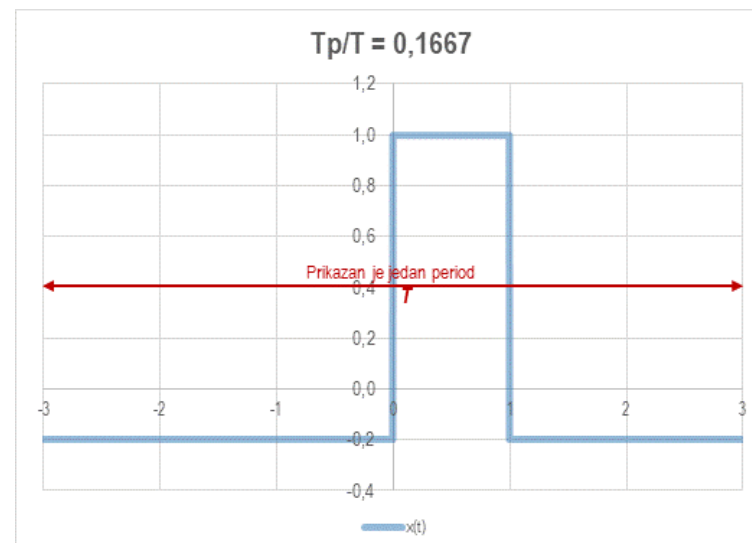
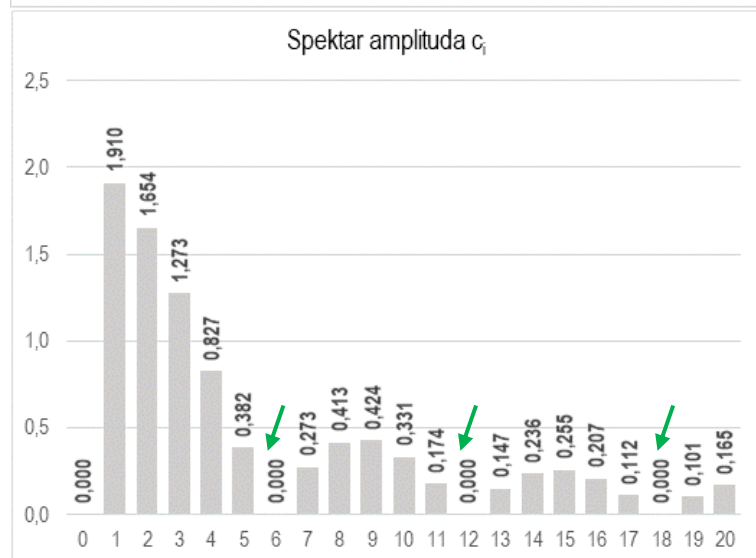
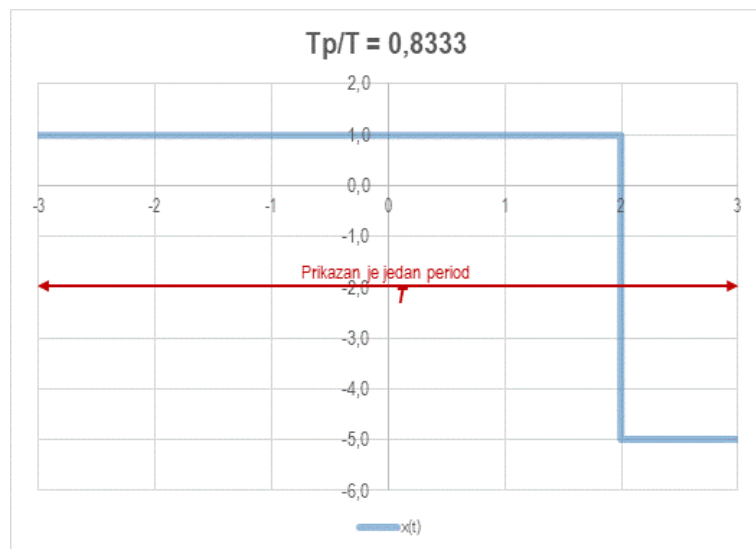
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8.



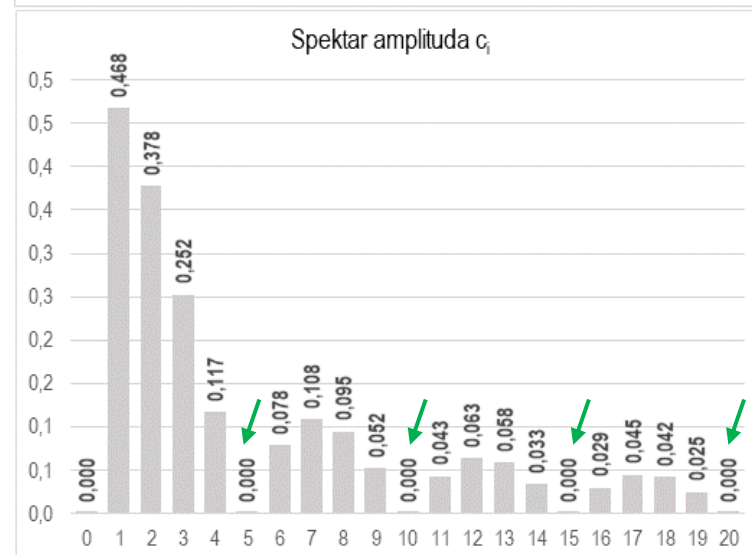
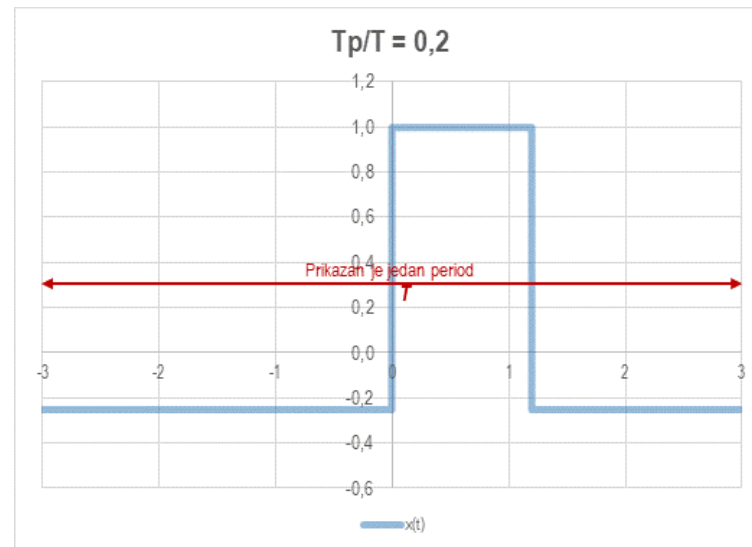
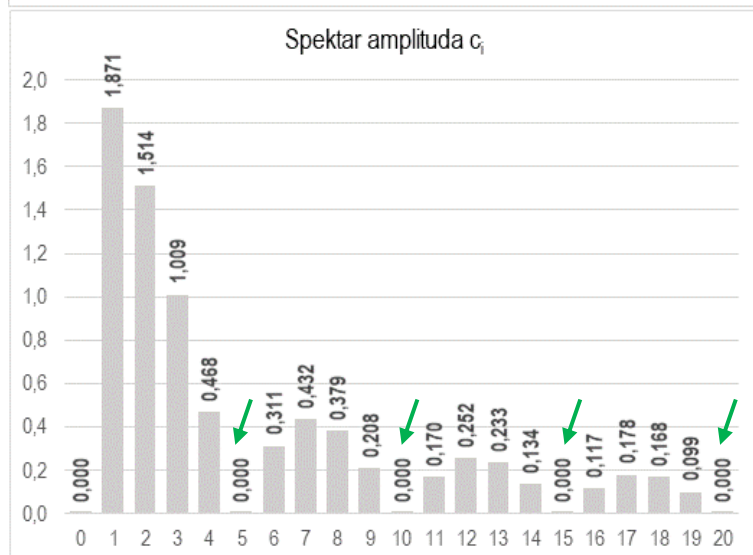
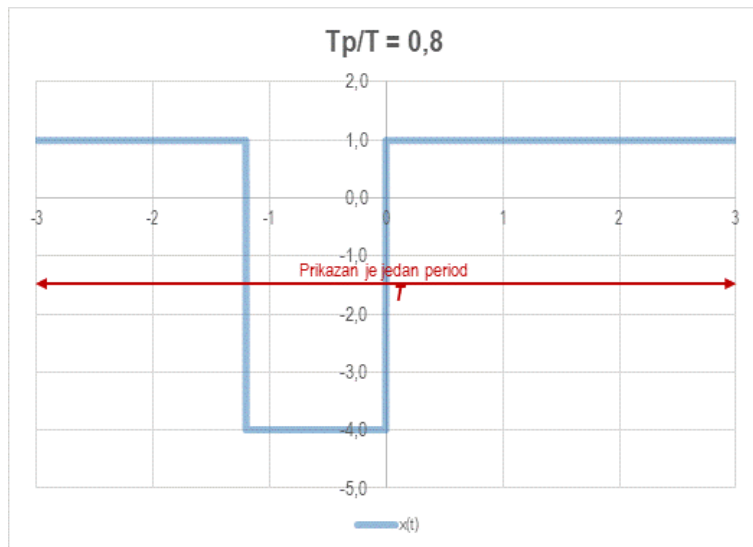
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Primjer 8.



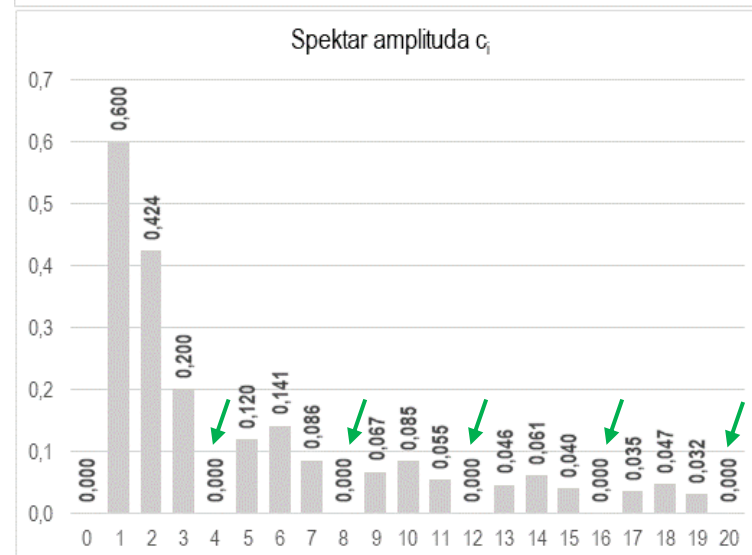
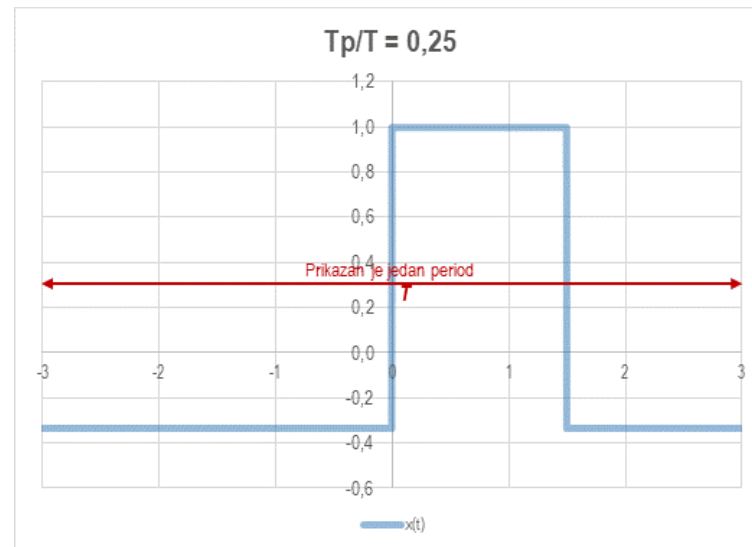
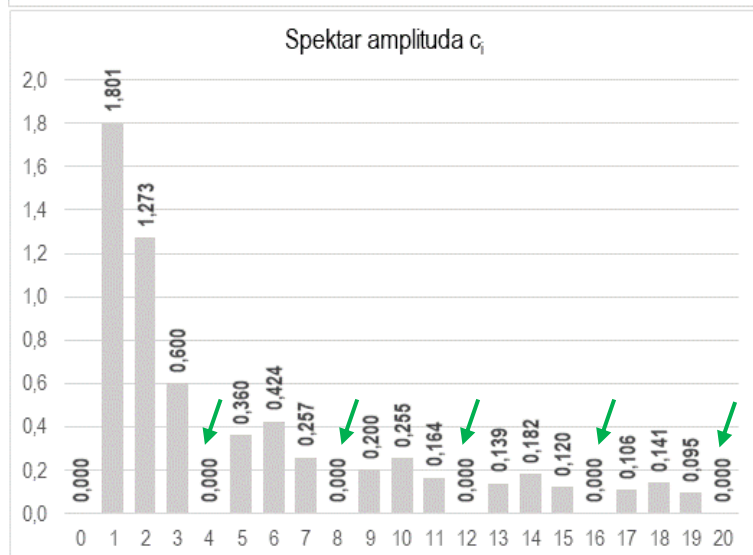
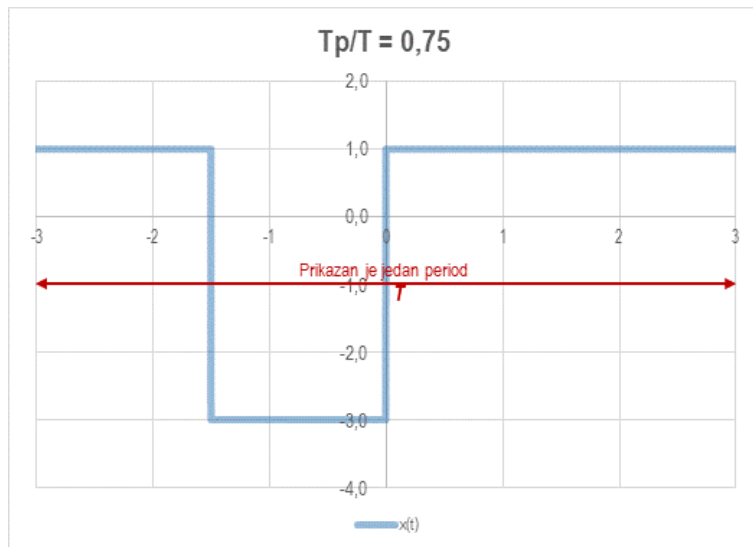
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Primjer 8.



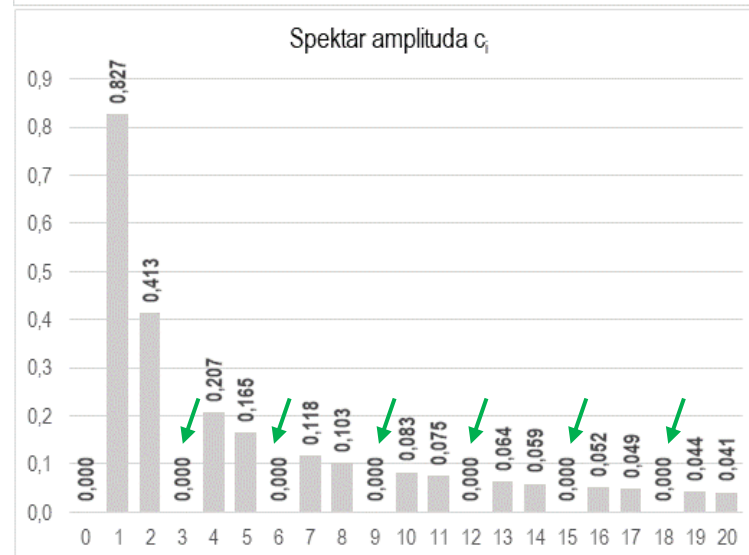
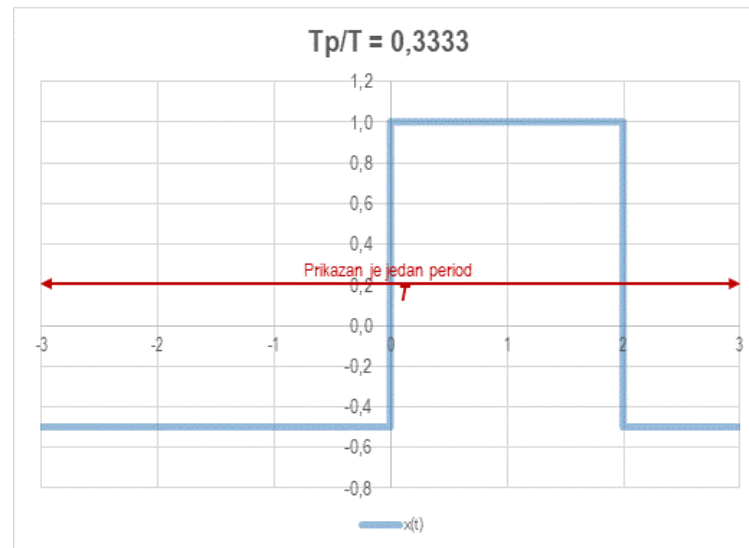
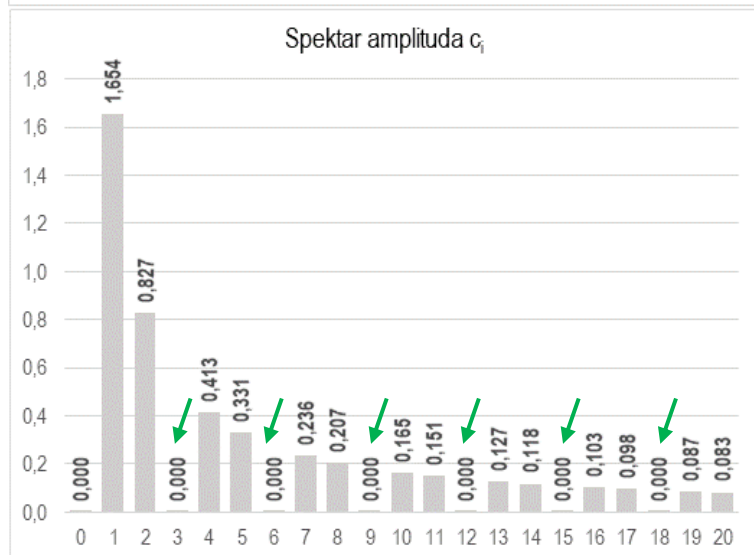
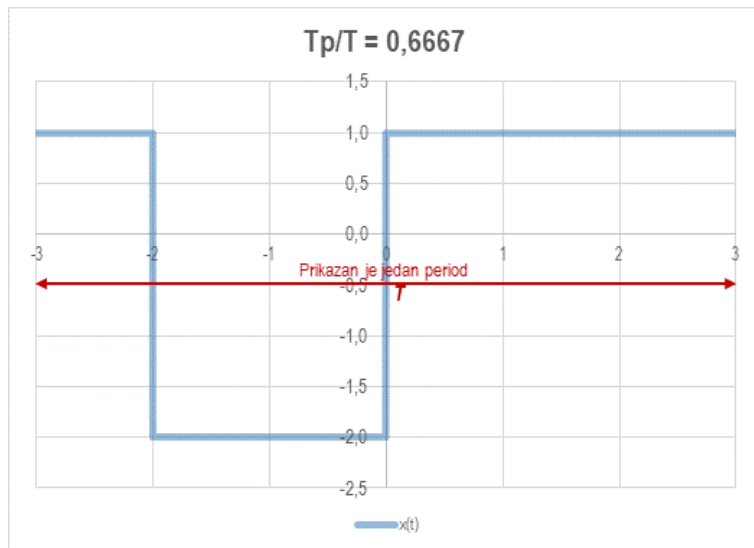
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Primjer 8.



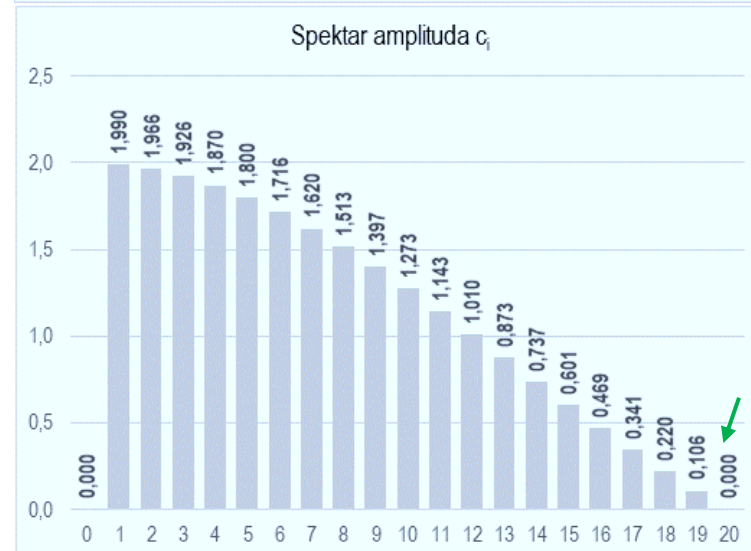
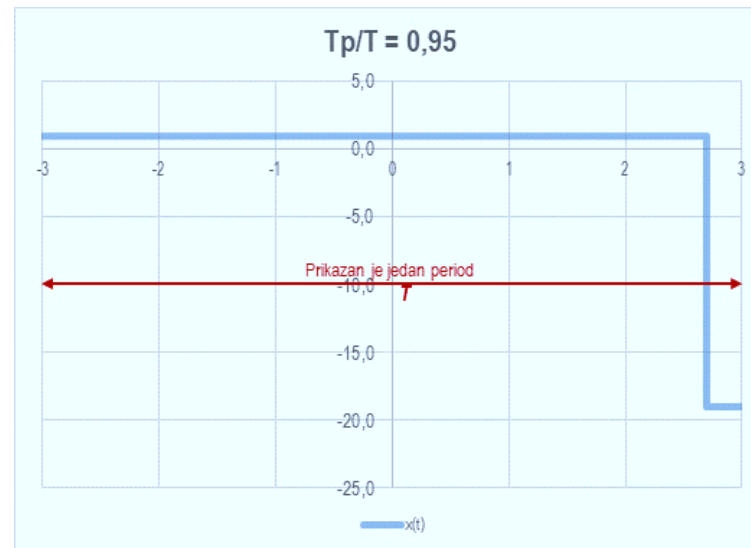
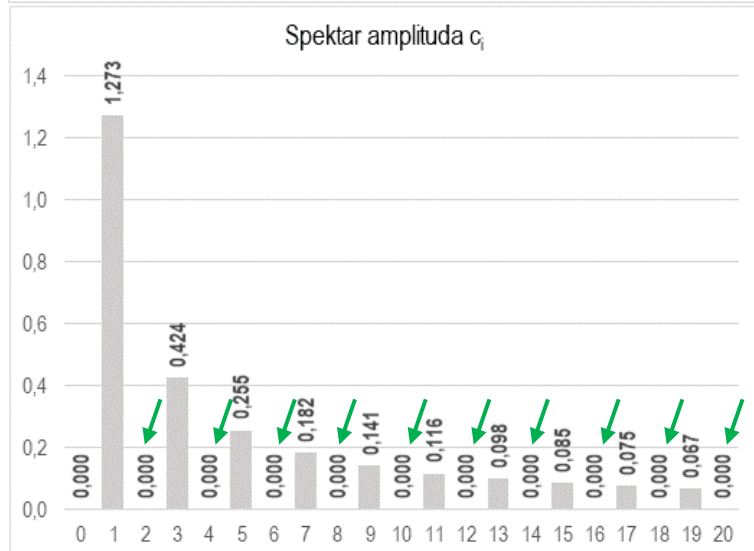
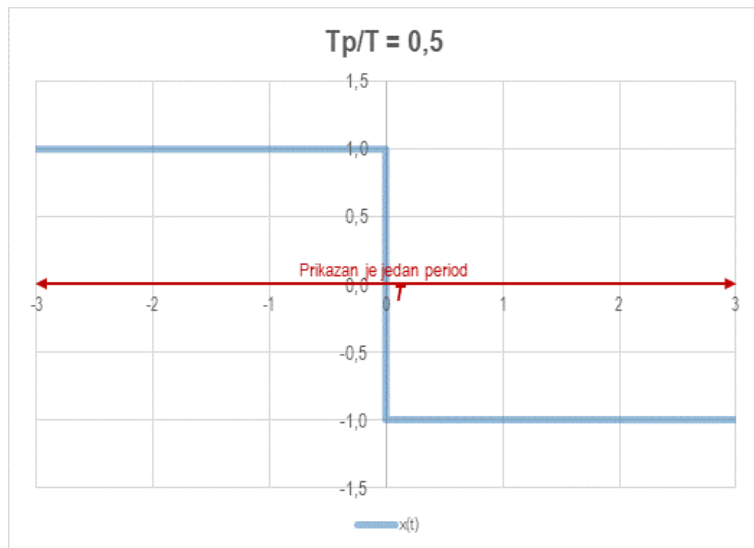
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Primjer 8.



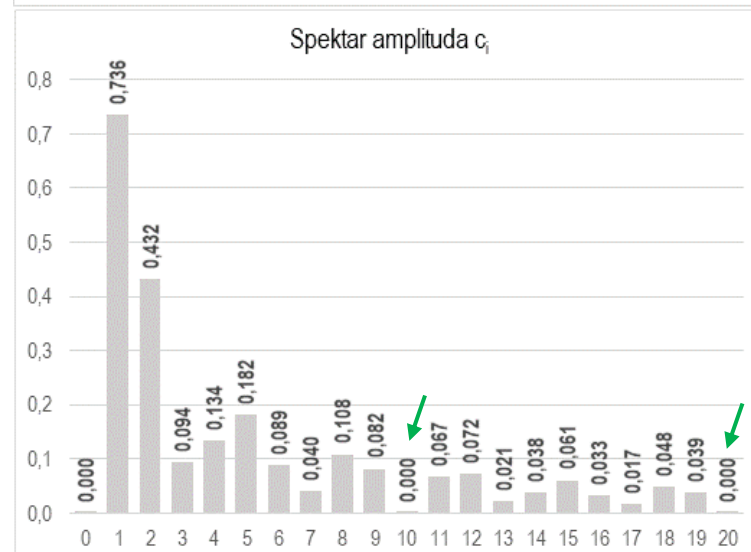
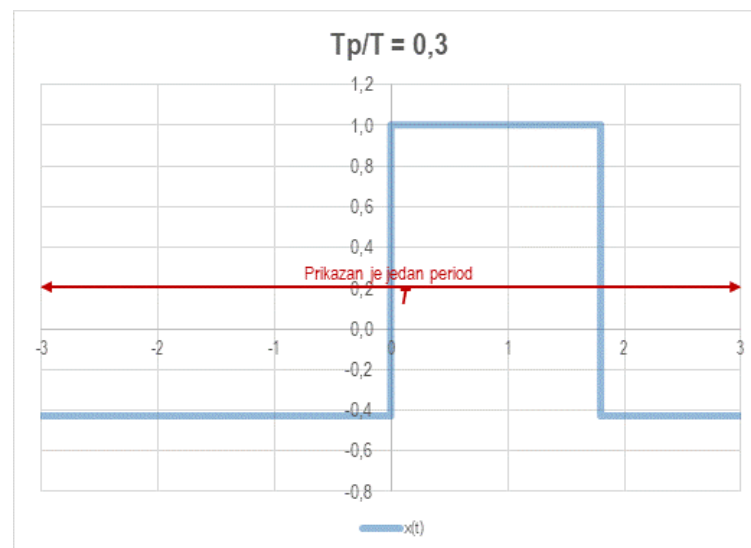
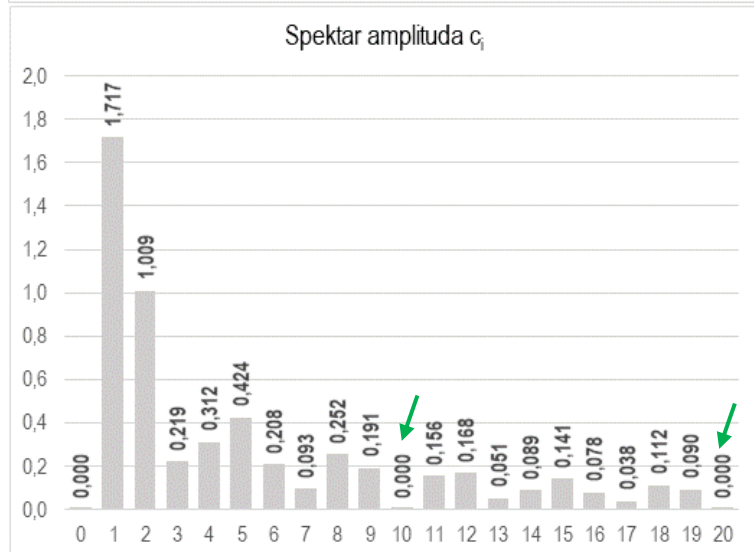
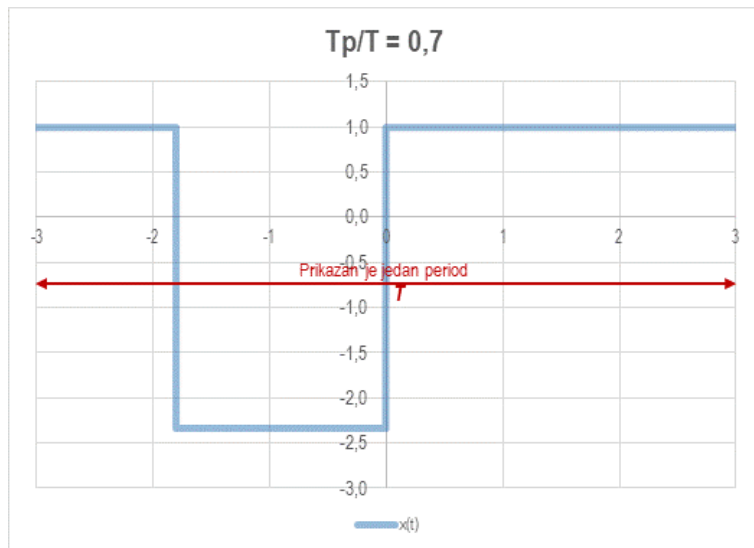
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8.



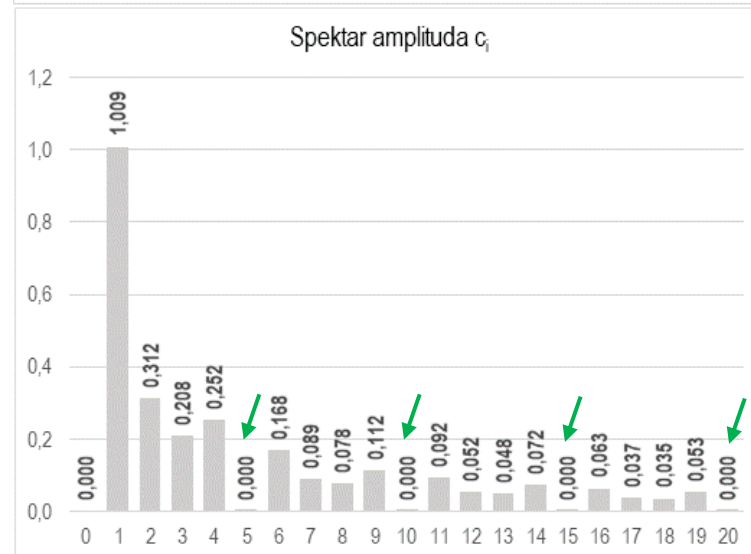
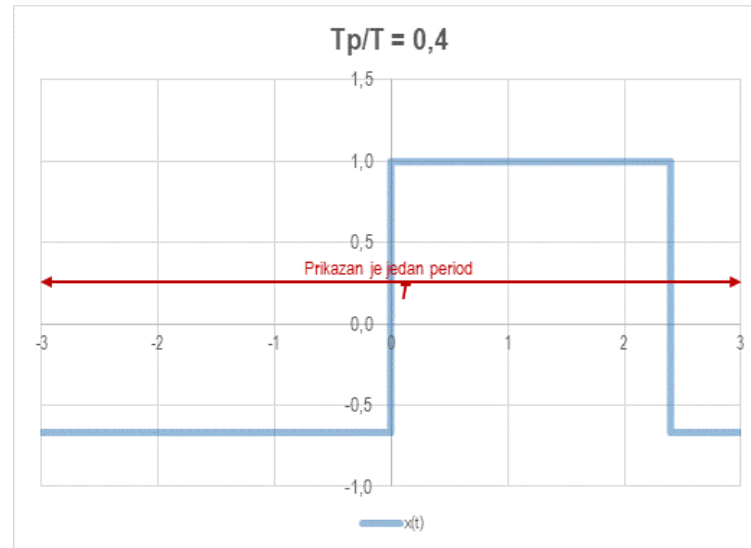
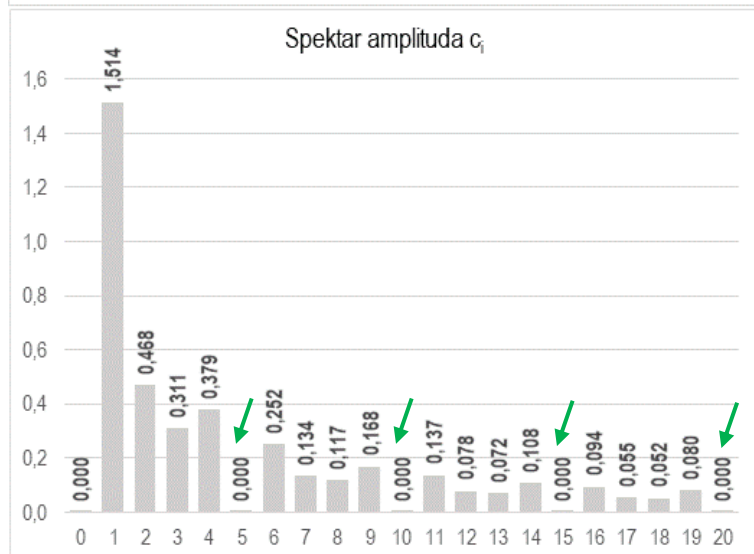
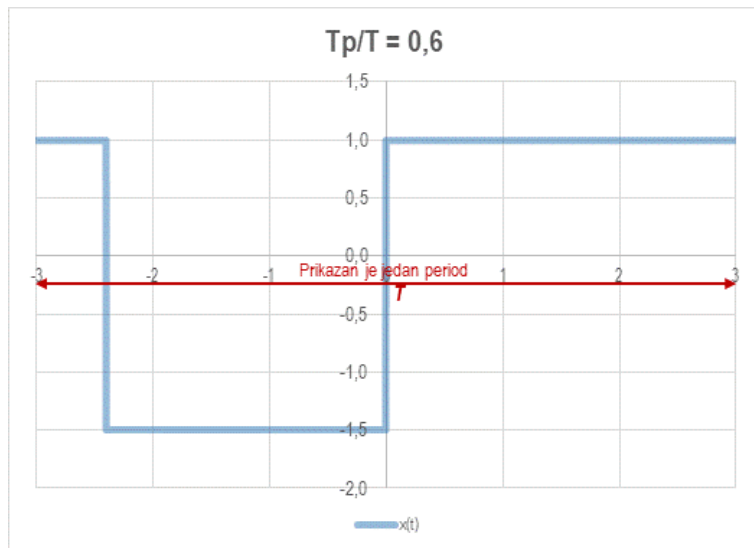
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8. - dodatak



Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Primjer 8. - dodatak



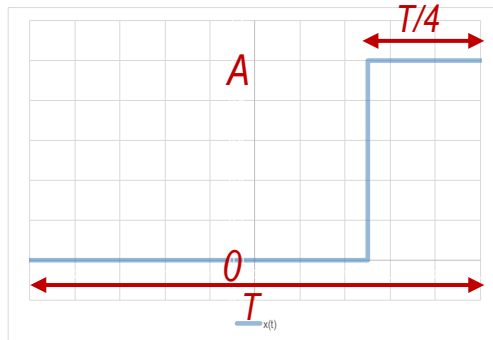
Moguće su manja odstupanja od teorijskih vrijednosti jer je korištena numerička integracija

Zadaci za vježbu (ZzV)



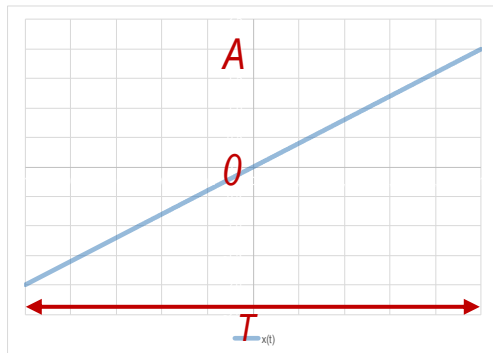
ZzV 09-01

Izvedite analitičke izraze za c_i , φ_i Fourierovog reda signala čiji je jedan period prikazan na slici:



ZzV 09-02

Izvedite analitičke izraze za c_i , φ_i Fourierovog reda signala čiji je jedan period prikazan na slici:



Primjeri: Animacija pomaka faznog spektra

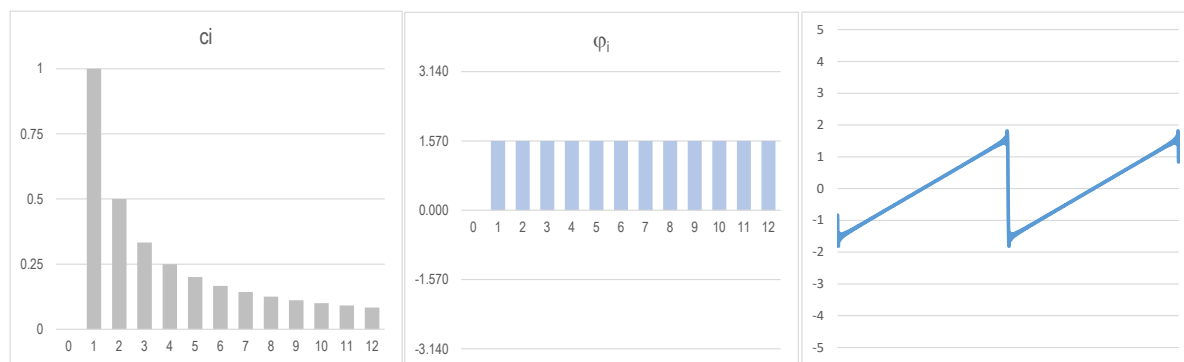


Immanuel Kant: „**Pojmovi su bez zorova prazni, a zorovi su bez pojmova slijepi**”

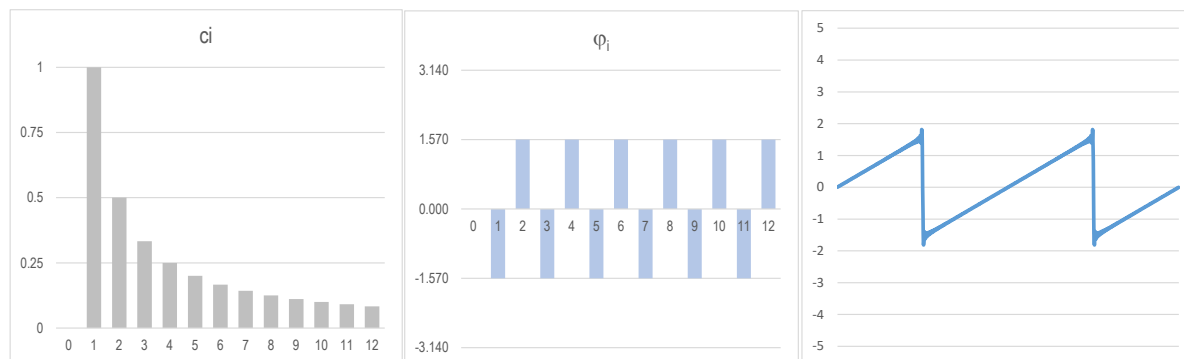
Sve animacije imaju isti amplitudni spektar: $c_i = 1/i$

Animacija sinteze signala provodi se promjenom faznog spektra: $\varphi_i \in [-\pi, \pi]$

Svi harmonici
nealternirajuća faza



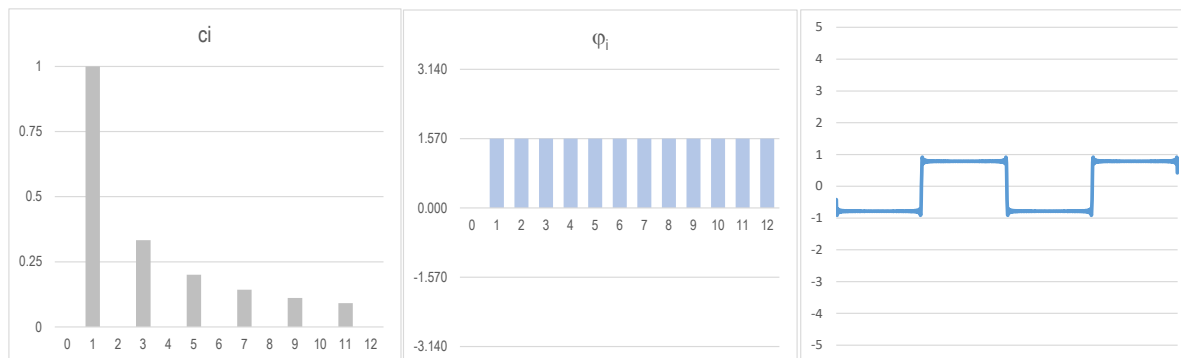
Svi harmonici
alternirajuća faza



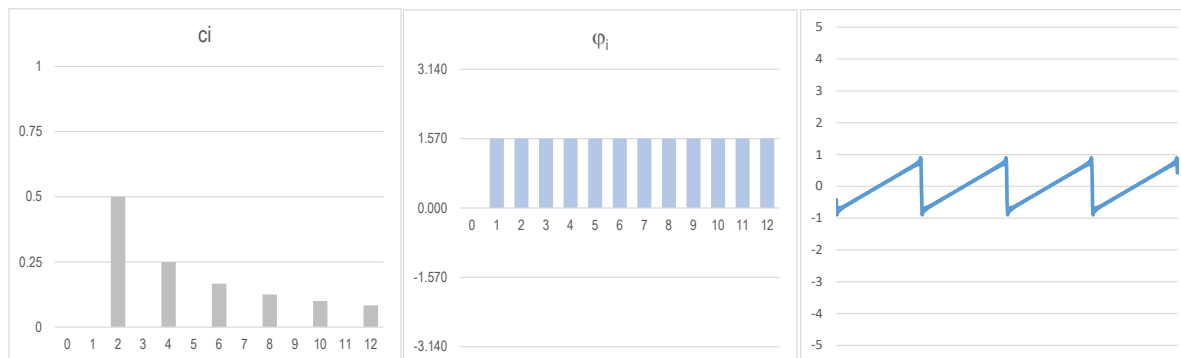
Primjeri: Animacija pomaka faznog spektra



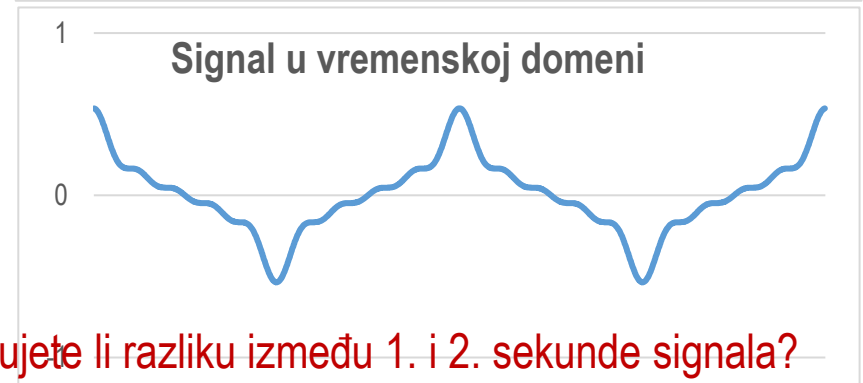
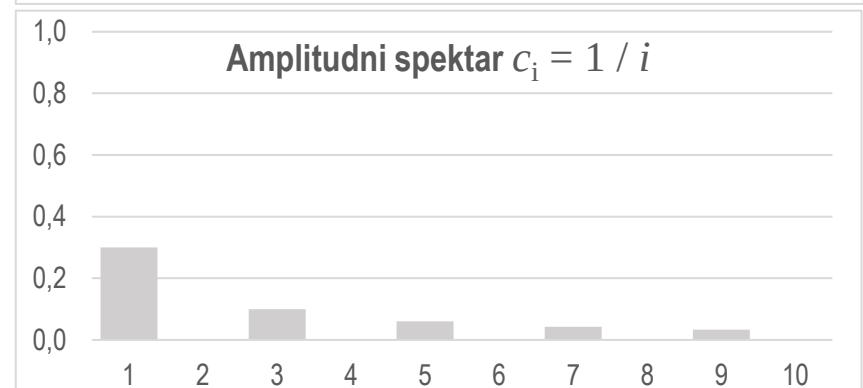
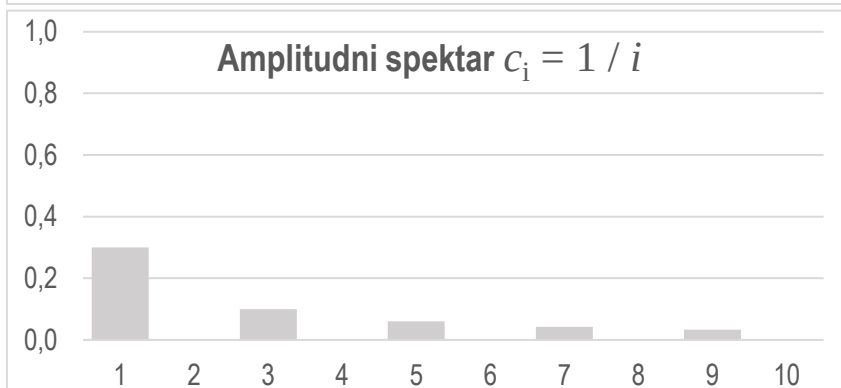
Neparni harmonici
nealternirajuća faza



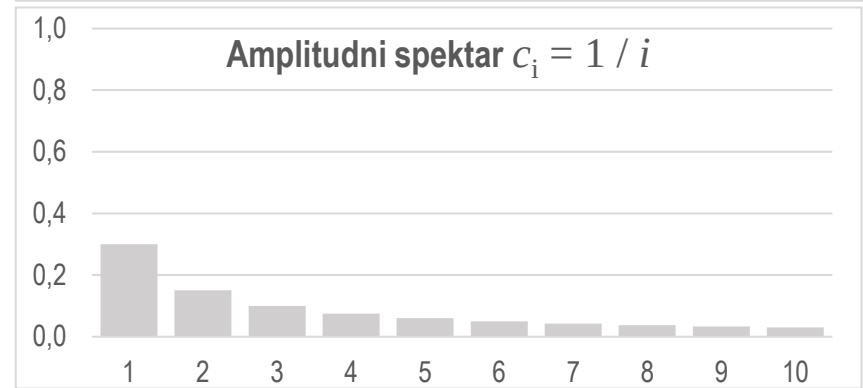
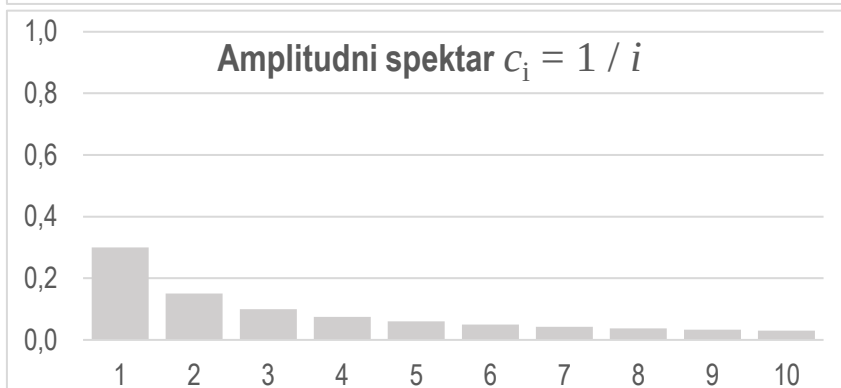
Parni harmonici
nealternirajuća faza



Audio primjer 1.: Faze pravokutnog signala



Audio primjer 2.: Faze pilastog signala



Prilog: Animacija FFT-a (za Dan ELO)

